

Project Report on
**SOLAR PANEL PERFORMANCE MONITORING AND
PREDICTIVE MAINTENANCE SYSTEM**

Submitted to Department of Industrial Engineering & Management

In partial fulfillment for the award of Degree of

Bachelor of Engineering
in
Industrial Engineering & Management

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DECEMBER - 2025

CERTIFICATE

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This is to certify that the Project Work Phase 2 (23IM7PCPW2)

Entitled

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ABSTRACT

Solar energy systems are increasingly being adopted as a sustainable alternative to conventional energy sources. However, solar panels experience degradation in performance over time due to various factors including dust accumulation, thermal stress, aging components, and technical faults. The absence of real-time monitoring systems in many solar installations leads to undetected efficiency losses, resulting in reduced energy generation and increased maintenance costs. This project presents the design and development of an intelligent Solar Panel Performance Monitoring and Predictive Maintenance System that continuously monitors solar panel parameters, detects performance anomalies, and provides predictive alerts before critical failures occur. The system employs IoT sensors, machine learning algorithms, and cloud-based analytics to ensure optimal solar panel performance and minimize downtime.

ACKNOWLEDGEMENT

We would like to thank the Department of Industrial Engineering and Management BMS College of Engineering, and the faculty of the department for giving us the opportunity to work on this topic for our sixth and seventh semester's major project.

We would like to express our gratitude to our guide Prof. Disha M Nayak, Assistant Professor, Department of Industrial Engineering and Management, BMS College of Engineering for guiding us throughout our project course and helping us in our major project.

We would like to express our sincere thanks to Dr. Shailaja V. N, Professor and Head of IEM Department, BMS College of Engineering, for providing a positive environment to make progress in our project.

We would like to express our sincere thanks to Prof. J Reddy Naik, Assistant Professor of IEM Department, BMS College of Engineering, for Co-ordinating us throughout the project.

We also thank our friend Syed Ali from Electrical and Electronics Engineering for helping us throughout the prototype development and electrical connections of the model.

We extend our gratitude to all the staff members of the Department of IEM, BMS College of Engineering, for their constant support and guidance.

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1. INTRODUCTION

Solar energy has emerged as one of the most promising sources of renewable power, providing a clean and sustainable alternative to fossil fuels. However, maintaining optimal efficiency in large-scale solar farms poses significant challenges due to panel degradation, environmental conditions, and undetected faults. Faults such as micro-cracks, dust accumulation, shading effects, and diode failures can reduce energy output and increase maintenance costs.

This research focuses on AI-driven fault detection and predictive analysis using IoT sensors, machine learning algorithms, and deep learning models. By leveraging real-time monitoring and predictive maintenance, the study aims to improve operational efficiency, minimize downtime, and enhance the overall performance of photovoltaic systems. The integration of AI technologies allows for automated diagnostics, ensuring maximum energy yield while reducing manual inspection efforts.

This project addresses the gap in solar panel monitoring by developing an intelligent system that combines real-time data acquisition, advanced analytics, and predictive modelling. The system aims to maximize energy output, reduce maintenance costs, and extend the operational life of solar panel installations through early fault detection and performance optimization.

1.1 Scope and Contribution

The scope of this work encompasses the design of sensor networks for parameter monitoring, development of data processing algorithms, implementation of machine learning models for predictive analytics, and creation of user-friendly interfaces for system management. The project also explores the integration of IoT technologies and cloud computing to enable remote monitoring and scalable deployment.

At the hardware level, the research involves implementing multi-parameter sensor networks capable of capturing electrical, thermal, and environmental data. The system utilizes sensors including ACS712 for current measurement, LM35 for panel temperature, DHT11 for ambient conditions, voltage sensors, and dust accumulation monitoring. These sensors interface with Arduino UNO and ESP8266 Wi-Fi modules for data acquisition and transmission.

The machine learning implementation leverages Random Forest algorithms for efficiency prediction, power output forecasting, and maintenance classification. Models are trained on

historical performance data and environmental measurements to establish baseline performance characteristics and fault signatures. The system uses MongoDB for data storage and FastAPI for backend services, with React-based frontend dashboards providing real-time visualization and control interfaces.

1.2 Methodology Overview

The research methodology follows a systematic approach encompassing system design, hardware prototyping, software development, model training, and validation. The data collection layer utilizes sensors to capture operational data including current (ACS712), voltage, panel temperature (LM35), ambient conditions (DHT11), and irradiance levels. Arduino UNO boards and ESP8266 Wi-Fi modules interface with sensors, perform data preprocessing, and transmit measurements to centralized storage systems.

Data transmission occurs through multiple pathways depending on deployment requirements. For prototyping, USB serial communication with PLX-DAQ software enables real-time logging into Excel spreadsheets. Production deployments utilize Wi-Fi connectivity through ESP8266 modules to transmit data to MongoDB cloud databases, chosen for flexible schema design and scalability for time-series sensor data.

The machine learning layer processes stored data using Random Forest algorithms for regression and classification tasks. Feature engineering transforms raw sensor readings into informative features including moving averages, time-based parameters, and derived efficiency metrics. Models undergo training with 80-20 train-test splits, cross-validation, and hyperparameter tuning. Trained models generate real-time predictions for efficiency forecasting, power output estimation, and maintenance requirement classification.

The intelligent analytics layer synthesizes model outputs to generate actionable insights including performance trend analysis, economic impact calculations, and predictive maintenance recommendations. The alert system implements graduated response strategies with dashboard notifications for minor variations, email alerts for medium-priority warnings, and SMS notifications for critical faults. User interfaces developed using React framework provide comprehensive dashboards with real-time KPIs, interactive historical charts, maintenance management modules, and system configuration options.

1.3 Overview of Problem

Solar panel systems face efficiency losses due to dust accumulation, temperature variations, shading effects, micro-cracks, and electrical faults that often go undetected. Traditional manual inspection methods are inadequate for large-scale installations, leading to reduced energy output, increased downtime, and higher maintenance costs. The absence of real-time monitoring systems results in reactive rather than proactive maintenance approaches.

1.4 Objectives to be achieved

To develop an intelligent IoT-based monitoring system that continuously tracks solar panel performance using sensors and machine learning algorithms. The system aims to detect faults early, predict maintenance requirements, optimize cleaning schedules, and maximize energy generation efficiency. It provides real-time dashboards for operators to make data-driven decisions and reduce operational costs through predictive maintenance strategies.

1.5 Expected Outcomes

This research demonstrates the feasibility of comprehensive, automated solar panel monitoring using cost-effective components suitable for retrofit installations. The system quantifies performance improvements through early fault detection and optimized maintenance scheduling. Economic benefits are assessed through cost-benefit analysis considering reduced downtime, extended equipment lifespan, and increased energy yield. The research contributes technical documentation, machine learning methodologies, and implementation guidance for intelligent renewable energy system management.

2. LITERATURE REVIEW

The field of solar panel monitoring has witnessed remarkable growth in recent years, driven by the increasing adoption of renewable energy systems and the need for efficient performance management. This literature review examines the current state of research in solar panel monitoring technologies, predictive maintenance strategies, and intelligent diagnostic systems.

2.1 Real-time Monitoring and IoT Technologies

Martinez et al. [1] conducted comprehensive research on real-time monitoring systems and demonstrated that continuous monitoring can significantly improve solar panel efficiency by detecting performance deviations early. Their findings suggest that well-implemented monitoring systems can enhance efficiency by identifying subtle changes that would otherwise go unnoticed.

Kumar and Singh [2] explored IoT integration with wireless sensor networks for solar panel monitoring. Their research highlighted the advantages of distributed sensor networks in capturing comprehensive performance data from large-scale installations. The authors demonstrated that IoT-based systems offer superior scalability compared to traditional wired solutions while addressing practical challenges like power consumption and data transmission reliability.

Wilson et al. [10] conducted extensive research on wireless communication protocols specifically tailored for solar energy systems, addressing unique requirements including long-range communication, power efficiency, and environmental robustness.

2.2 Machine Learning and Predictive Analytics

Chen et al. [3] investigated various machine learning algorithms for solar panel performance prediction, comparing regression models, decision trees, and neural networks. Their analysis revealed that machine learning techniques can accurately forecast performance under varying environmental conditions and identify degradation patterns.

Hassan et al. [16] provided broader perspective on AI applications in renewable energy systems, exploring how AI technologies can transform the entire ecosystem from individual component monitoring to grid-level optimization.

2.3 Advanced Diagnostic and Communication Systems

Thompson and Lee [4] developed innovative diagnostic techniques for comprehensive fault detection, focusing on identifying specific faults including partial shading, soiling, and cell degradation. Their multi-parameter approach combines electrical measurements with thermal imaging for accurate fault classification.

Liu et al. [8] advanced the field by investigating data fusion techniques, developing algorithms that process multiple sensor inputs to create holistic views of solar panel health. Their work showed that data fusion approaches significantly improve diagnostic accuracy compared to single-sensor systems.

Rodriguez et al. [5] explored edge computing for real-time solar monitoring, demonstrating how local processing capabilities can reduce bandwidth requirements while improving system responsiveness and reliability.

2.4 Economic Optimization and Maintenance Strategies

Johnson et al. [9] conducted comprehensive cost-benefit analysis demonstrating that monitoring systems provide substantial return on investment through improved energy generation and reduced maintenance costs. Their economic models help stakeholders evaluate financial benefits of monitoring solutions.

Turner et al. [14] focused on economic optimization of maintenance scheduling, developing algorithms that consider component reliability, maintenance costs, and energy losses to determine optimal schedules. White et al. [11] expanded this concept by investigating predictive maintenance strategies using real-time performance data.

2.5 System Design and Regulatory Considerations

Anderson et al. [7] investigated optimal sensor placement strategies, developing mathematical models to determine configurations that maximize monitoring effectiveness while minimizing costs. Their work provided practical guidelines for cost-effective network design.

Park and Kim [6] explored weather forecasting integration, demonstrating that predictive models incorporating meteorological data can improve forecasting accuracy by 15-20% and enable proactive system adjustments.

Roberts et al. [15] addressed the regulatory landscape, examining existing standards and identifying gaps in current frameworks. Evans et al. [13] investigated cybersecurity considerations, identifying vulnerabilities and proposing protection strategies for IoT-based systems.

Adams et al. [12] conducted environmental impact assessment of monitoring technologies, examining lifecycle effects and providing guidelines for sustainable monitoring solutions.

2.6 Research Gaps and Future Directions

Current literature indicates needs for more sophisticated fault diagnosis algorithms and monitoring systems that can adapt to different installation types without extensive customization. The integration of AI with monitoring systems shows promise, but more research is needed for algorithms that operate effectively with limited training data. Economic optimization requires continued investigation, particularly for small and medium scale installations where cost constraints are stringent

3. PROBLEM FORMULATION

The problem addressed in this project is the lack of intelligent, real-time monitoring and maintenance systems for solar panels, which leads to undetected performance degradation and energy losses. Solar installations often operate with issues such as dust accumulation, temperature-induced stress, and electrical faults that are not identified promptly in the absence of continuous diagnostics. This creates a gap between the theoretical efficiency of photovoltaic systems and their actual field performance, increasing lifecycle costs and reducing reliability. Therefore, there is a need to formulate a system that continuously monitors critical parameters, detects anomalies early, and enables predictive maintenance to ensure optimal solar panel performance

3.1 PROBLEM DEFINITION

Solar energy has emerged as one of the most reliable and sustainable renewable energy sources for meeting the growing global demand for electricity. Photovoltaic (PV) systems are widely deployed in residential, commercial, and industrial applications due to their environmental benefits and long-term cost advantages. However, the performance and efficiency of solar panels deteriorate over time due to several unavoidable operational and environmental factors. These include dust accumulation on panel surfaces, elevated operating temperatures, humidity variations, partial shading, aging of components, and minor electrical or mechanical faults.

In most conventional solar installations, especially in small- and medium-scale setups, there is a lack of continuous and automated performance monitoring mechanisms. The performance of solar panels is often assessed through periodic manual inspections or by observing significant drops in power output. Such reactive maintenance practices result in delayed identification of faults and inefficiencies. Minor issues that initially cause marginal performance loss may remain undetected for long durations and eventually lead to substantial energy losses, increased downtime, and accelerated degradation of solar panels.

Furthermore, the absence of real-time monitoring prevents effective analysis of key electrical and environmental parameters such as voltage, current, power output, panel temperature, ambient conditions, and dust levels. Without continuous data collection and analysis, it becomes difficult to differentiate between normal performance variations caused by environmental conditions and abnormal behavior resulting from faults or degradation. This lack of visibility limits the ability of system operators to take timely

corrective actions and optimize overall system performance.

Therefore, there exists a significant need for an integrated and intelligent solar panel performance monitoring system capable of real-time data acquisition, continuous performance evaluation, and early fault detection. Such a system should be able to analyze historical and real-time data, identify performance deviations, and generate predictive maintenance alerts before critical failures occur. Addressing this problem is essential to enhance energy generation efficiency, minimize operational losses, extend the lifespan of solar panels, and ensure the reliable operation of solar power systems.

3.2 OBJECTIVES

The main objective of this project is to design and implement an intelligent solar panel performance monitoring and predictive maintenance system that maximizes energy output while minimizing downtime and maintenance costs. This is achieved by integrating IoT-based data acquisition, machine learning models, and user-friendly analytics to continuously evaluate panel health and efficiency.

Performance monitoring is an important objective focused on continuously tracking key parameters such as voltage, current, power, temperature, humidity, dust level, and irradiance to assess real-time operating conditions. By capturing high-quality, time-stamped data from a multi-sensor array, the system ensures accurate visibility into the performance of the solar panels.

Predictive maintenance is a core objective aimed at shifting from reactive or time-based maintenance to condition-based and predictive interventions. Using machine learning models like Random Forest and anomaly detection techniques such as Isolation Forest, the system forecasts faults, cleaning needs, and degradation trends before they become critical.

Intelligent analytics is a key objective that focuses on transforming raw data into actionable insights for decision-makers. Through dashboards, trend analysis, efficiency evaluation, and economic metrics such as ROI and cost of power loss, the system supports informed operational and maintenance planning.

User-centric system design is an essential objective that ensures the solution is practical and deployable in real-world solar installations. This includes developing a responsive web-based dashboard, configurable thresholds and alerts, maintenance management features.

4. METHODOLOGY

The flowchart-based methodology explains how the system starts with multi-sensor data collection from the solar panel, followed by edge preprocessing where noise is filtered, values are validated, and key performance metrics like power are computed. The processed data then flows to the cloud or storage, where machine learning models perform performance prediction, anomaly detection, and maintenance forecasting to identify issues early.

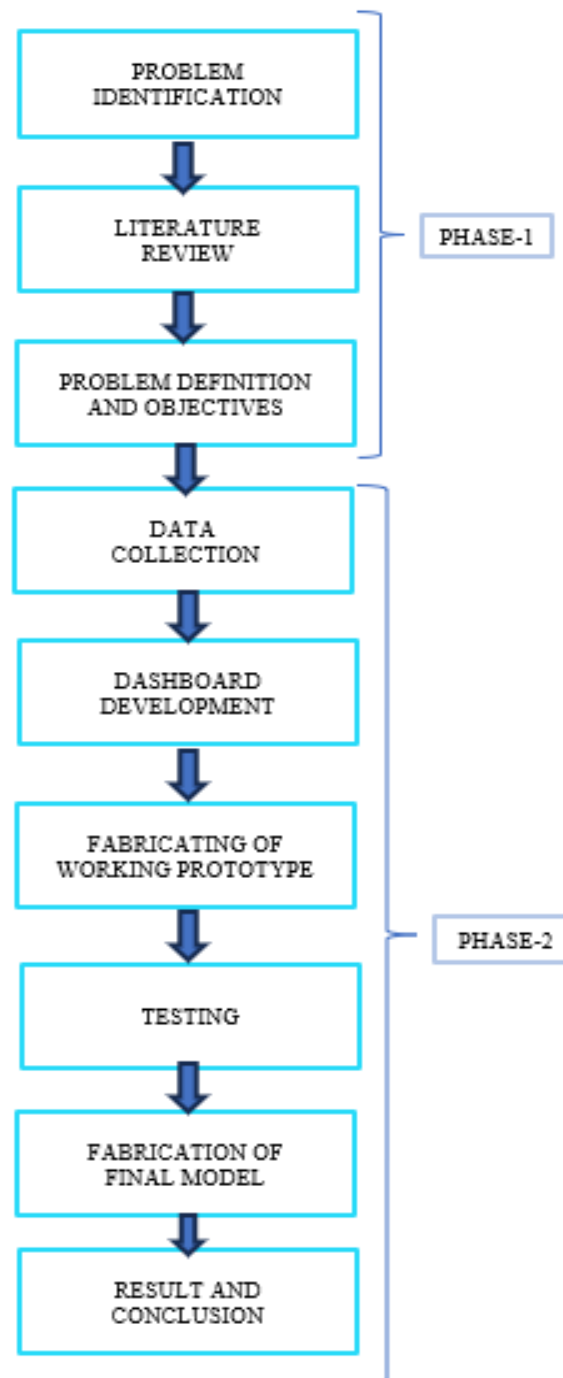


Fig. 4.1 Flowchart of Methodology

4.1 Layers of the Model

The methodology for the “Solar Panel Performance Monitoring and Predictive Maintenance System” is designed using a multi-layer intelligent architecture, aligned with modern IoT– AI based industrial system

1. Data Collection Layer
2. Data Processing Layer
3. Machine Learning & Analytics Layer
4. Intelligent Analytics Layer
5. Alert & Response System
6. User Interface & Output Layer

Each layer plays a critical role in capturing solar data, processing it in real time, applying predictive models, generating intelligent insights, and enabling user-friendly monitoring for maintenance and decision making.

4.1.1 Data Collection Layer

This is the foundation of the system. All raw environmental, electrical, and performance parameters are collected from multiple sensors.

Multi-Sensor Array

The following sensors were installed and interfaced with Arduino Uno & WI-FI Module:

Table 4.1.1. Multi sensor Array

Sensors	Parameters	Purpose
ACS712	Current	Power measurement (W)
LM35	Panel temperature	Identifying overheating / hotspots
DHT11	Ambient temp & humidity	Weather influence on performance
25V Voltage sensor	DC voltage sensing (optional expansion)	Grid / inverter monitoring
LM2596 Buck Converter	Power regulation (from 9v to 5v)	Stable supply to components
Irradiance sensor	Intensity of rays	To collect the irradiance data
Dust Sensor	Percentage of dust on the panel	To detect the effect of dust on panel

Collected Parameters

- Voltage (V)
- Current (A)
- Power ($P = VI$)
- Panel temperature ($^{\circ}\text{C}$)
- Ambient temperature ($^{\circ}\text{C}$)
- Humidity (%)
- Dust Accumulation (%)
- Irradiance (W/m^2)

Thermal Imaging (Future-ready layer)

While LM35 measures temperature, future versions may include:

- IR thermal camera
- Hotspot detection
- Crack detection

4.1.2 Edge Processing Layer

As soon as data is collected, local processing occurs before cloud transmission.

Data Preprocessing

The following preprocessing steps were performed:

- Noise filtering using moving average
- Validation of analog readings
- Removal of extreme spikes
- Conversion of raw ADC values into engineering units
- Local timestamp addition
- Faulty/missing values corrected using interpolation

Edge Analytics

Executed directly inside Arduino / Node MCU:

- Real-time computation of power ($P = V \times I$)
- Threshold-based anomaly detection
- Comparing panel temperature vs ambient
- Checking efficiency drop conditions
- Detecting zero-current situations

Data Transmission

Processed data is transmitted to:

- MS Excel (PLX-DAQ – v2.11) via Node MCU (Wi-Fi Module)
- Virtual pins carry: voltage, current, power, temperature, humidity, Irradiance
- Optimized transmission interval (2–5 seconds)
- Reduced bandwidth using filtered data.

4.1.3 Machine Learning & Analytics Layer

Cloud-stored data undergoes ML-based analysis, improving the intelligence of the solar system.

Performance Prediction

Using:

- Random Forest
- LSTM (future upgrade)

Parameters used for prediction:

- Voltage
- Current
- Panel Temperature
- Humidity

- Time of day

Model Output:

- This helps track expected vs actual performance.

Fault Detection

Implemented using Isolation Forest Algorithm:

It detects:

- Dust accumulation
- Temperature abnormality
- Sudden current drop
- Shading
- Wiring faults

Maintenance Prediction

Predicts:

- Cleaning schedule
- Fault likelihood
- Panel degradation rate
- Efficiency loss trends

This transforms monitoring → predictive maintenance.

4.1.4 Intelligent Analytics Layer

High-level analytics allow users and technicians to interpret long-term trends.

Performance Analytics

Includes:

- Power generation curve
- Efficiency trend
- Hourly/daily variance
- Output drop pattern
- Comparison with predicted values

Economic Analysis

Helps calculate:

- Cost of power loss
- ROI (Return on Investment)
- Payback period
- Cleaning cost vs recovered power

Predictive Insights

Model generates:

- “Clean your panel in 2 days”
- “Humidity rise may reduce panel output”

These insights reduce human supervision.

4.1.5 Alert & Response System

The system supports automated monitoring with real-time alerting.

Multi-Level Alerts

Alerts include:

- Low Power Alert
- Over Temperature Alert
- Efficiency Loss Alert
- Sensor Fault Alert
- Zero Current Alert

Automated Actions

Possible automated corrective actions:

- Send alert to user via dashboard
- Activate cleaning (future expansion)
- Switch load or battery backup
- Trigger thermal shutdown

Notification System (Future expansion)

software sends:

- SMS
- Email
- In-app notifications
- Real-time IoT dashboard alerts.

4.1.6 User Interface & Output Layer

The final layer presents data to the user in a clear and actionable manner.

Real-Time Dashboard

Displays:

- Voltage
- Current
- Power
- Panel Temperature
- Ambient Conditions
- Fault Status
- Efficiency Graph

Graph types:

Line graph, Bar graph, Predicted vs Actual graph

Reports & Analytics

Provides:

- Weekly summary
- Monthly performance report
- Sensor logs
- Fault occurrence log
- Cost savings report

5. SYSTEM ARCHITECTURE

The system architecture is designed as a multi-layer IoT–AI framework that connects field-level sensors and Arduino-based edge processing with cloud-side machine learning, analytics, and data storage services. It integrates a FastAPI–MongoDB backend with a React-based frontend dashboard, enabling seamless data flow from sensor acquisition to real-time visualization, alerting, and predictive maintenance insights.

5.1 TECHNOLOGY STACK

Multiple technologies have been used in frontend, backend and database for developing this predictive model dashboard.

5.1.1 Frontend Technologies

- Framework: React 19.0.0
- Routing: React Router DOM 7.5.1
- UI Components: Radix UI component library
- Styling: Tailwind CSS 3.4.17
- Charts: Recharts 3.1.2
- HTTP Client: Axios 1.8.4
- Form Management: React Hook Form 7.56.2
- Build Tool: CRACO (Create React App Configuration Override)

5.1.2 Backend Technologies

- Framework: FastAPI 0.110.1
- Server: Unicorn 0.25.0
- Database: MongoDB (Motor async driver 3.3.1)
- Machine Learning: scikit-learn $\geq 1.3.0$, NumPy $\geq 1.26.0$, Pandas $\geq 2.2.0$
- Model Persistence: Joblib $\geq 1.3.0$
- Environment Management: python-dotenv $\geq 1.0.1$

5.1.3 Machine Learning Models

- Efficiency Prediction Model: Random Forest Regressor
- Power Output Prediction Model: Random Forest Regressor
- Maintenance Classification Model: Random Forest Classifier
- Feature Scaling: Standard Scaler

5.2 FEATURES AND FUNCTIONALITY

The system offers comprehensive features and functionality including real-time performance monitoring, predictive analytics, maintenance management, configurable alerts, and rich dashboard visualizations to support data-driven solar panel operation and upkeep.

5.2.1 Dashboard Module

The main dashboard Fig 5.1 and Fig 5.2 provides a comprehensive overview of the solar panel system:

Key Performance Indicators (KPIs)

- Current Power Output (kW): Real-time power generation monitoring with trend indicators
- System Efficiency (%): Overall system efficiency calculation with historical comparisons
- Panel Temperature (°C): Real-time temperature monitoring with optimal range indicators (25-40°C)
- Dust Level Index: Categorized as Low/Medium/High with color-coded status indicators

Visualization Components

- Power Output Trend Chart: 24-hour power generation curve
- Efficiency Analysis Chart: System efficiency over time
- Efficiency Distribution: Pie chart showing performance breakdown
- Maintenance Status Panel: Overview of panel health
- Recent Alerts: Real-time system notifications

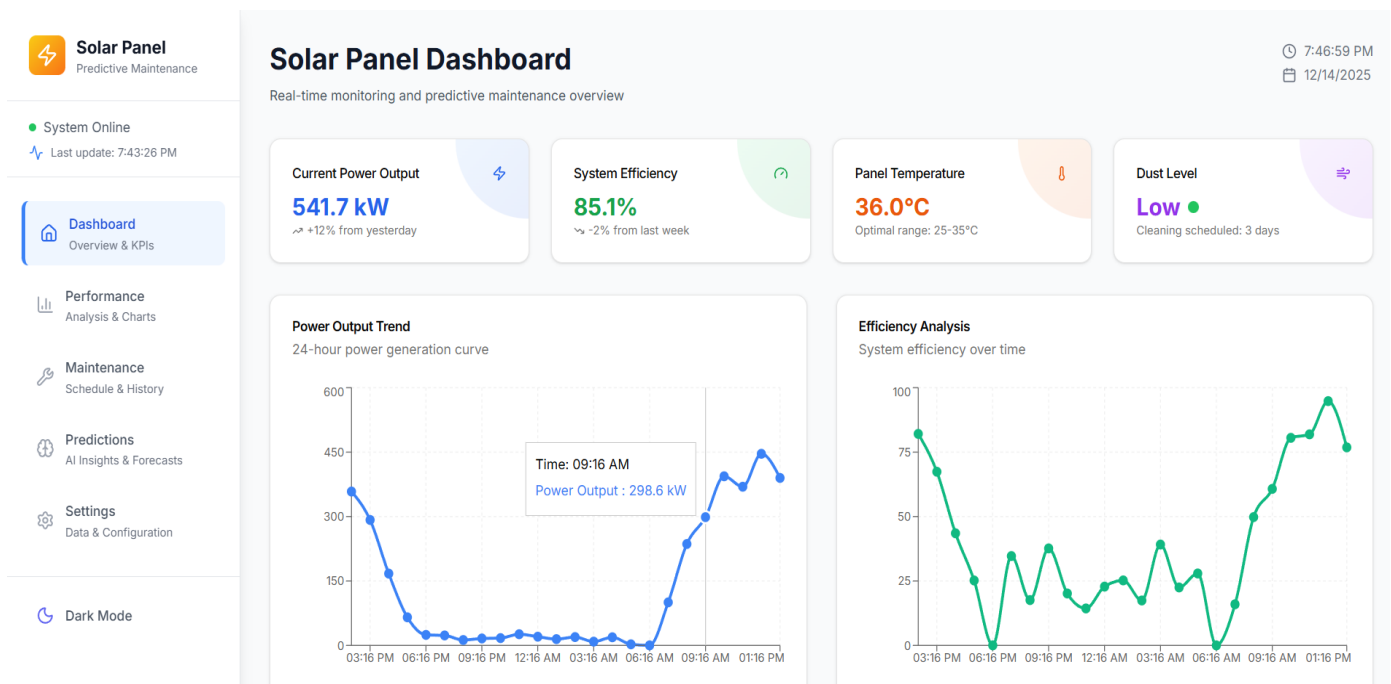


Fig. 5.1 Solar Panel Dashboard - 1

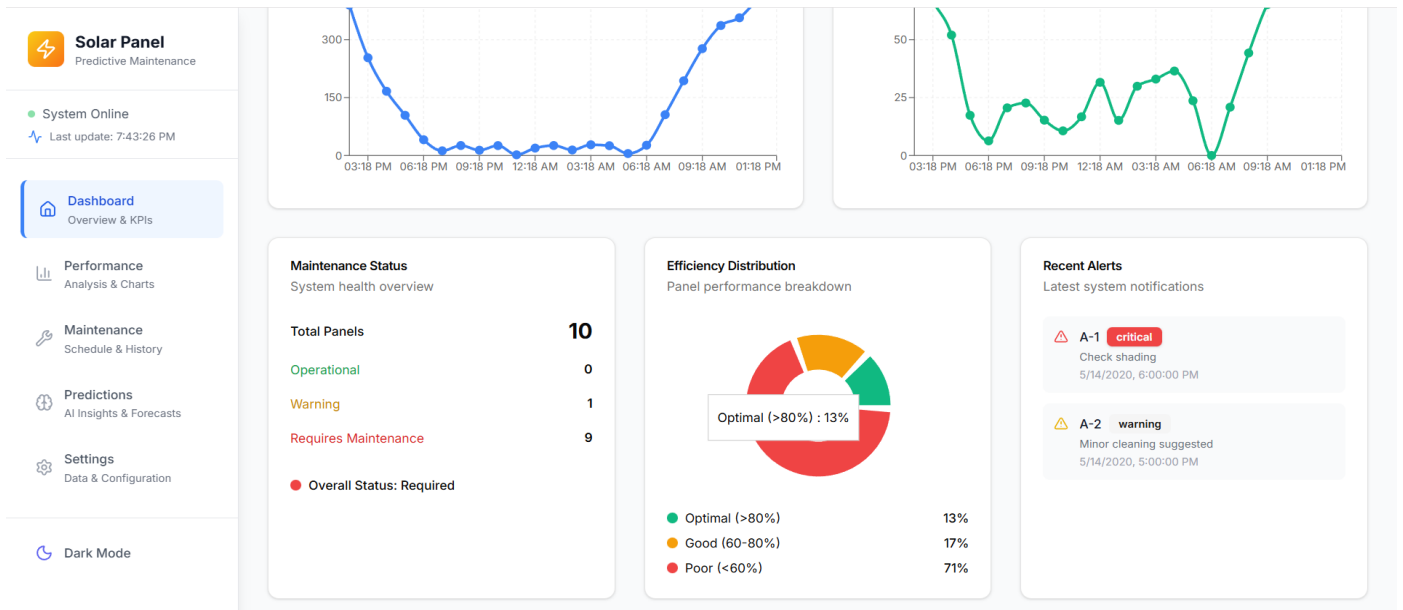


Fig. 5.2. Solar Panel Dashboard - 2

5.2.2 Performance Module

Detailed performance analytics and historical data visualization is shown in Fig 5.3, Fig 5.4 and Fig 5.5

- Multi-day performance analysis (1, 7, 30, 90 days)
- Power output vs. expected power comparison
- Efficiency trends and patterns
- Environmental factor correlation (temperature, humidity, irradiance)
- Performance deviation analysis
- Downloadable performance reports (PDF)

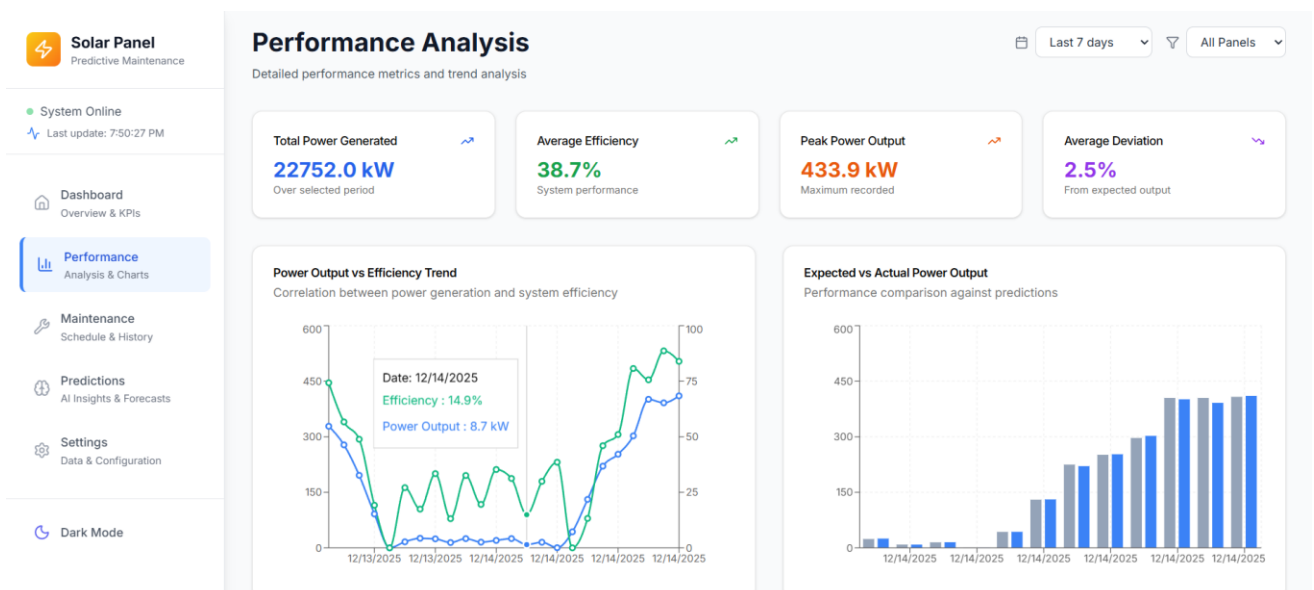


Fig. 5.3. Performance Analysis - 1

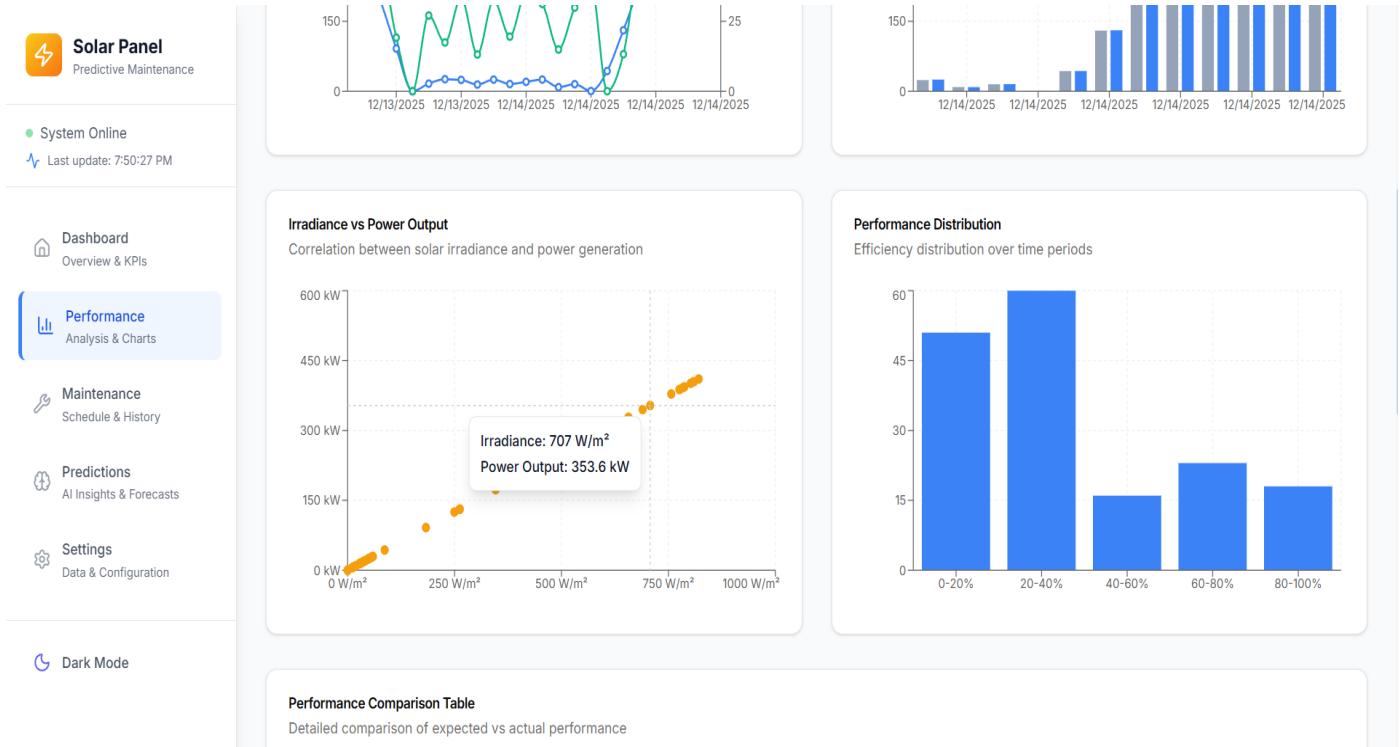


Fig. 5.4. Performance Analysis - 2

System Online

Last update: 7:50:27 PM

Dashboard

Overview & KPIs

Performance

Analysis & Charts

Maintenance

Schedule & History

Predictions

AI Insights & Forecasts

Settings

Data & Configuration

Dark Mode

Performance Comparison Table

Detailed comparison of expected vs actual performance

Time	Expected Output	Actual Output	Deviation	Efficiency	Status
12/13/2025	-30.3 kW	0.0 kW	0.0%	0.0%	Normal
12/13/2025	15.8 kW	16.3 kW	+3.0%	27.1%	Normal
12/13/2025	27.0 kW	25.8 kW	-4.5%	17.4%	Normal
12/13/2025	23.4 kW	24.3 kW	+3.9%	33.4%	Normal
12/13/2025	14.0 kW	14.3 kW	+2.2%	13.1%	Normal
12/13/2025	24.5 kW	24.9 kW	+1.6%	32.6%	Normal
12/14/2025	15.2 kW	15.4 kW	+1.2%	19.5%	Normal
12/14/2025	20.8 kW	20.1 kW	-3.2%	35.3%	Normal
12/14/2025	23.9 kW	25.0 kW	+4.7%	31.2%	Normal
12/14/2025	8.9 kW	8.7 kW	-1.9%	14.9%	Normal
12/14/2025	14.9 kW	15.2 kW	+2.0%	29.9%	Normal
12/14/2025	0.4 kW	0.4 kW	+3.1%	38.5%	Normal

Fig 5.5. Performance Comparison Table

5.2.3 Maintenance Module

Comprehensive maintenance management system is as shown in Fig 5.6:

- Maintenance task scheduling
- Task priority management (Low, Medium, High, Critical)
- Task status tracking (Scheduled, In Progress, Completed, Cancelled)
- Assigned technician management
- Task filtering and search
- Automated maintenance recommendations
- Task completion tracking

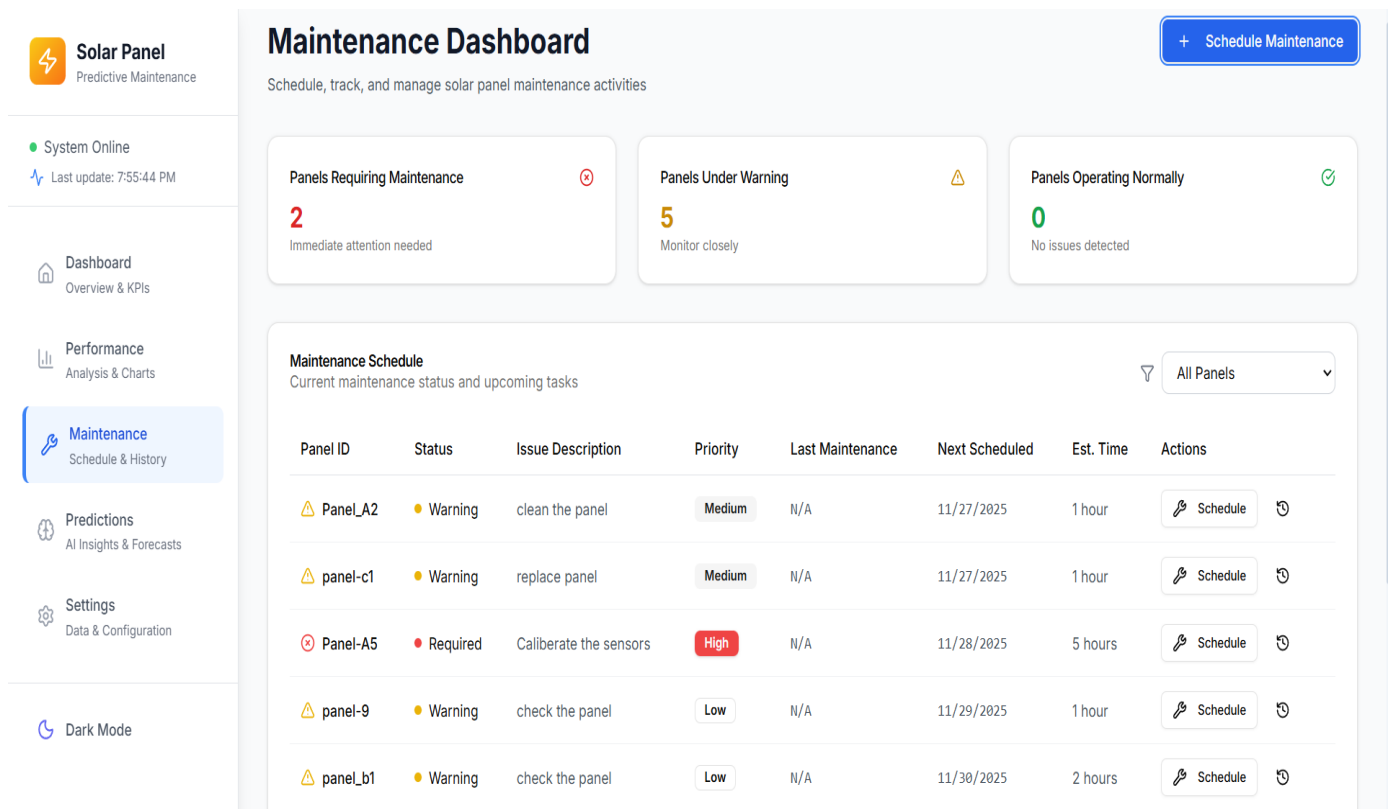


Fig. 5.6. Maintenance page

5.2.4 Predictions Module

Machine learning-powered predictive analytics is shown in Fig 5.7 and Fig 5.8:

- Power Output Forecast (7 Days): Predicted power generation with confidence intervals
- Efficiency Forecast: Future efficiency trends and degradation predictions
- Maintenance Predictions: Predicted maintenance requirements with risk probability
- AI-Driven Recommendations: Data-driven maintenance suggestions and optimization tips
- And performance of the prediction module is shown in Fig. 5.9.

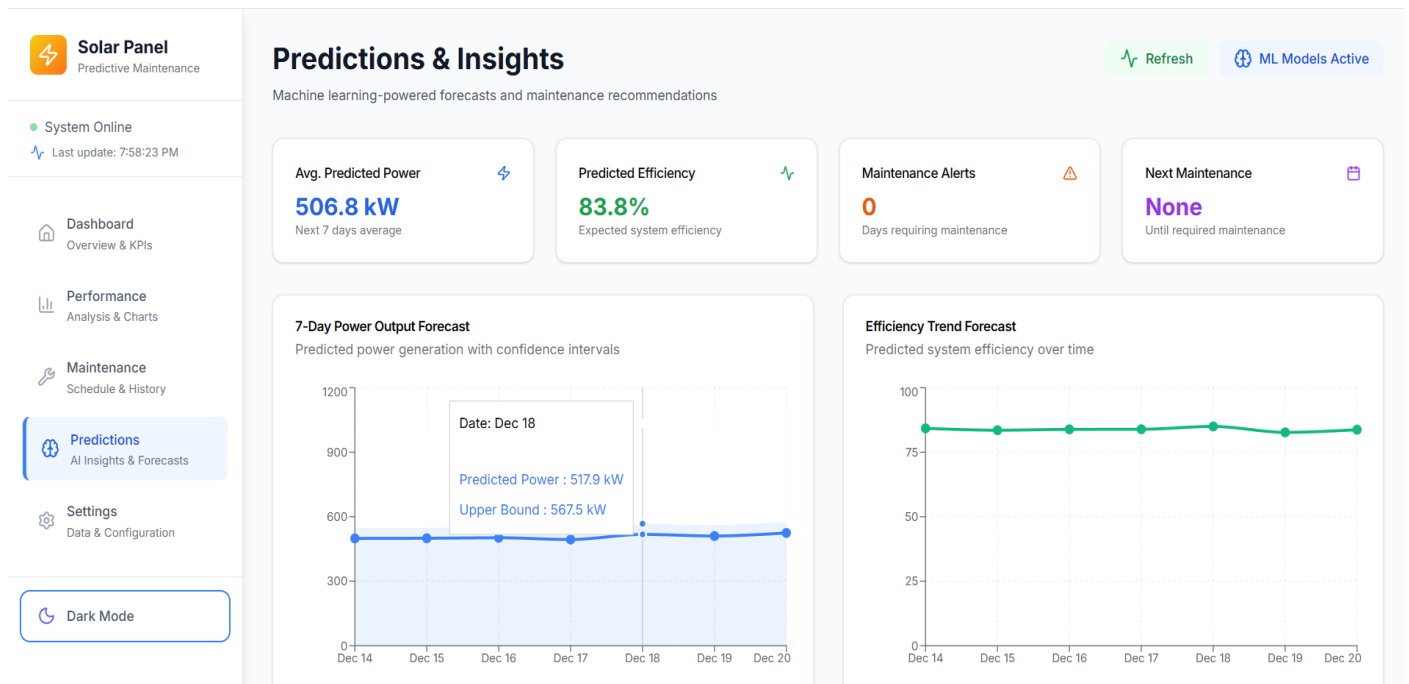


Fig. 5.7. Predictions and Insights - 1

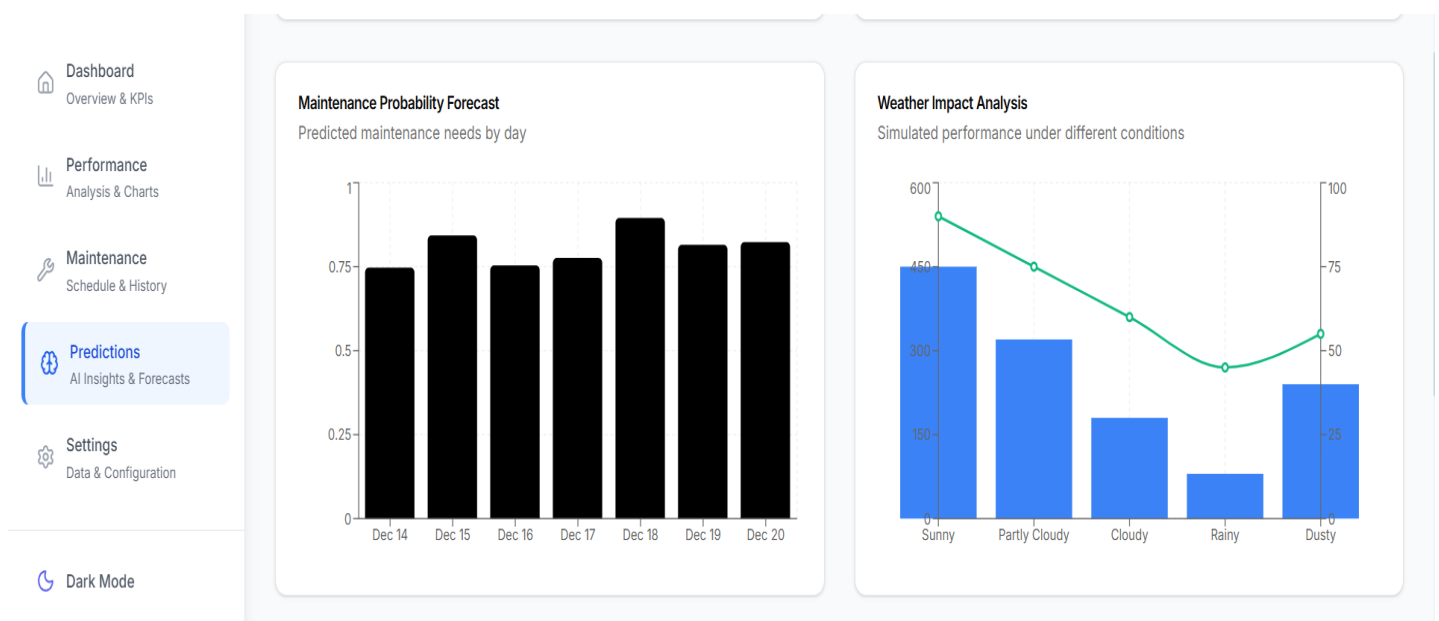


Fig. 5.8. Prediction and Insights - 2

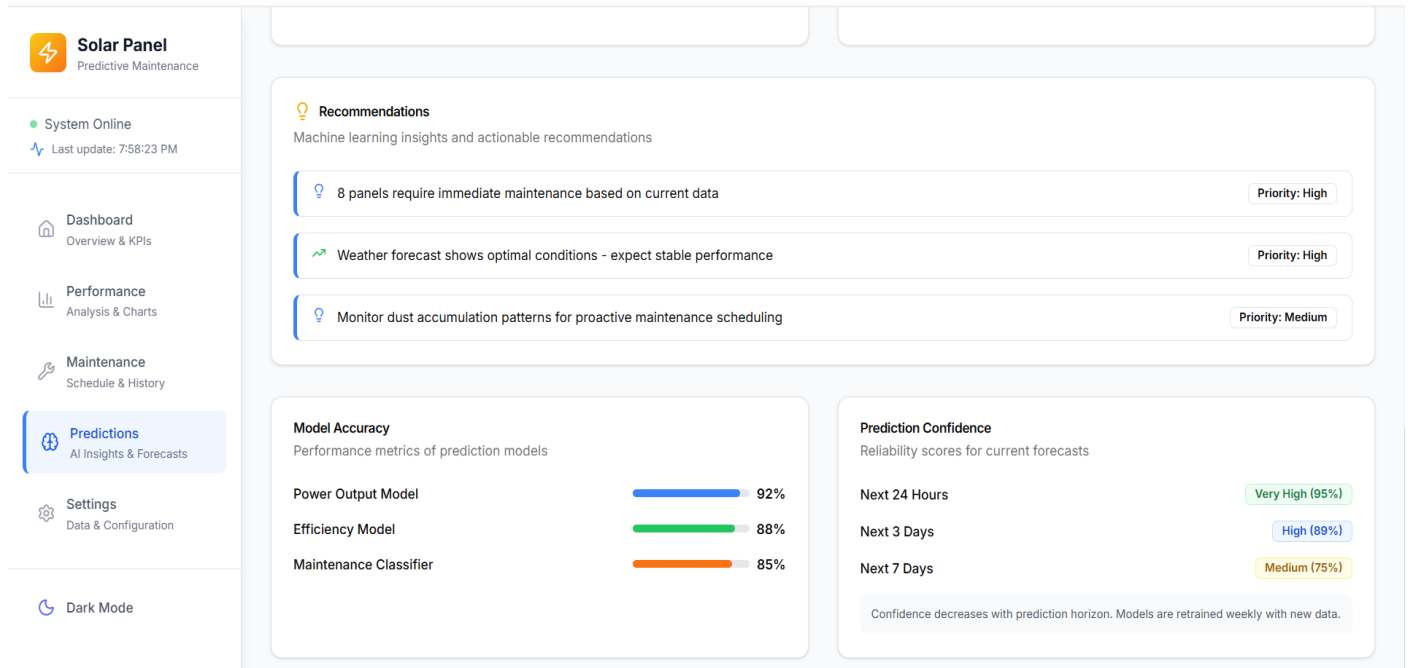


Fig. 5.9. Prediction Model Performance

5.2.5 Settings Module

There are multiple pages in setting module which is shown in Fig.5.10, 5.11, 5.12 and Fig 5.13.

- User management (Create, Read, Update, Delete)
- Role-based access control (Admin, Technician, Viewer)
- System threshold configuration
- Alert settings customization
- Data import/export functionality
- CSV data upload for model training
- System preferences

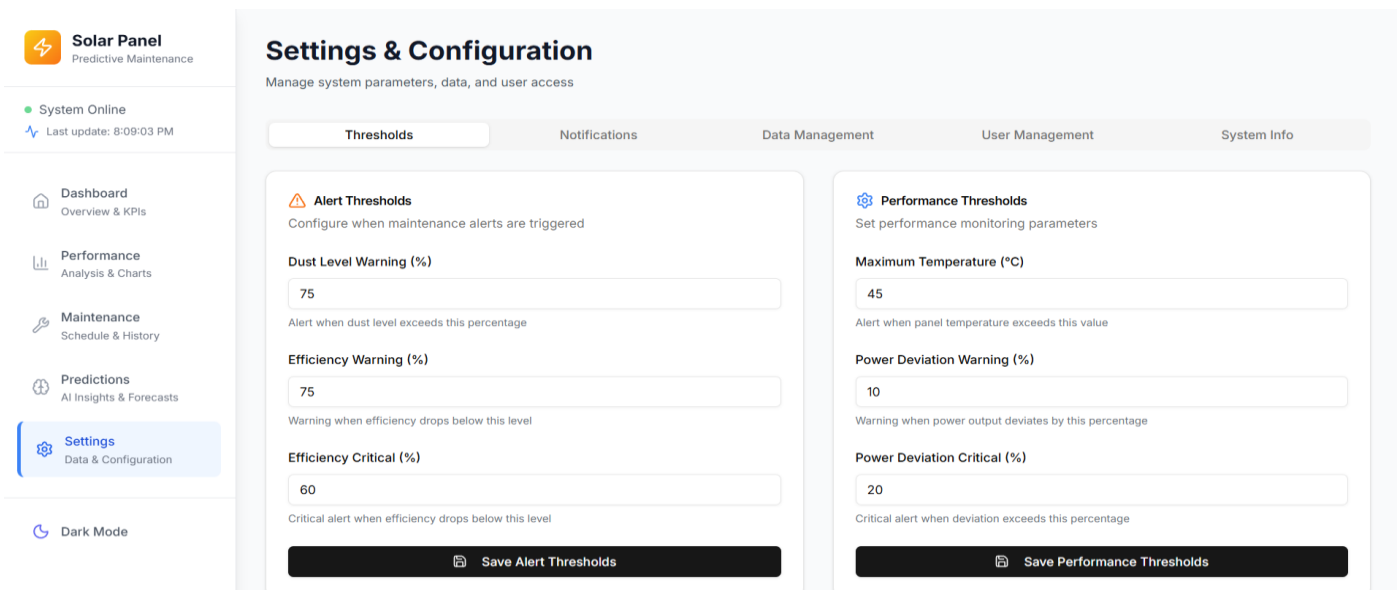


Fig.5.10. Thresholds settings

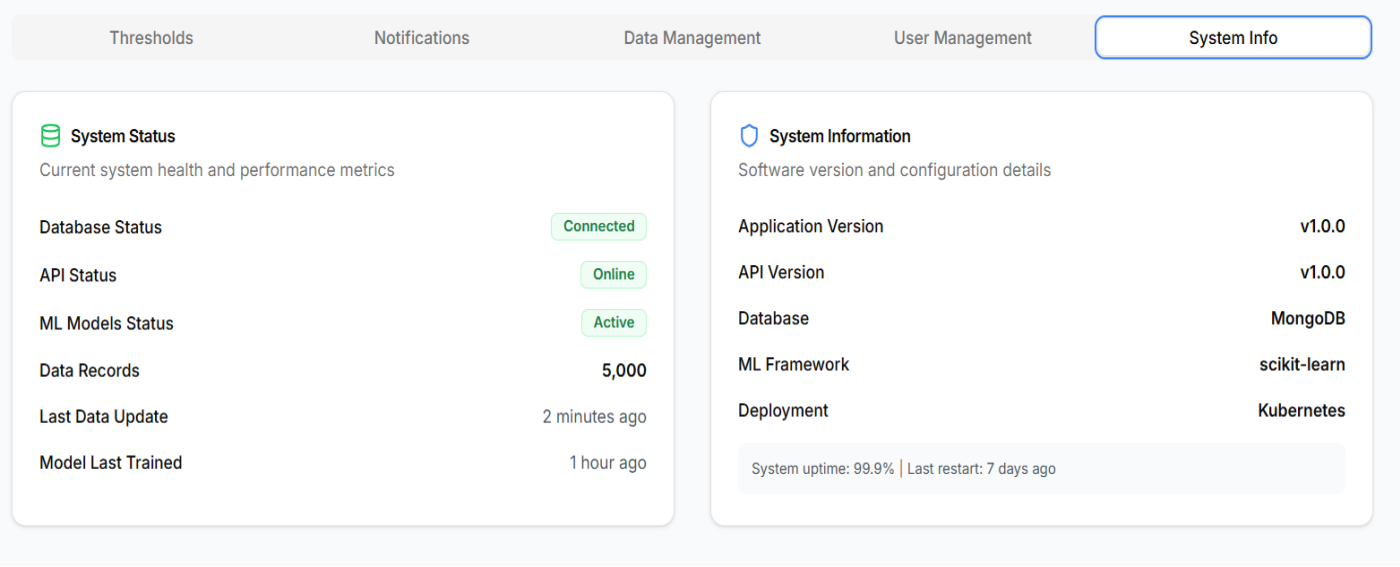


Fig. 5.11. User Management

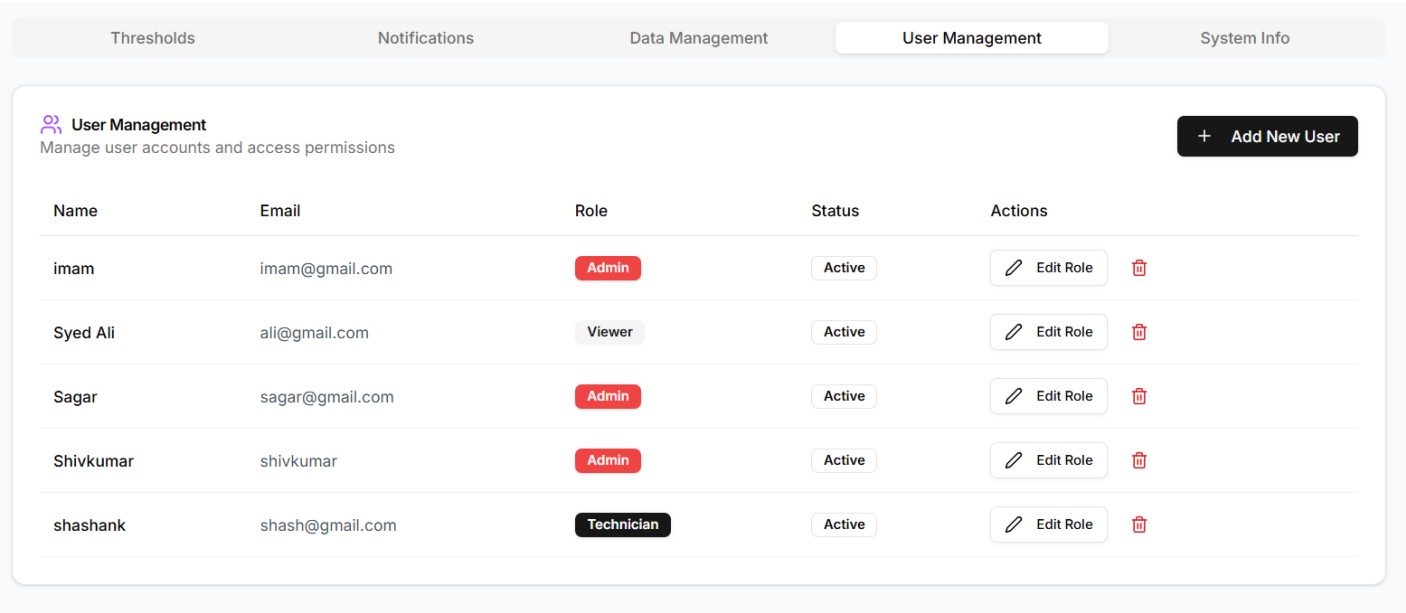


Fig.5.12. System Information

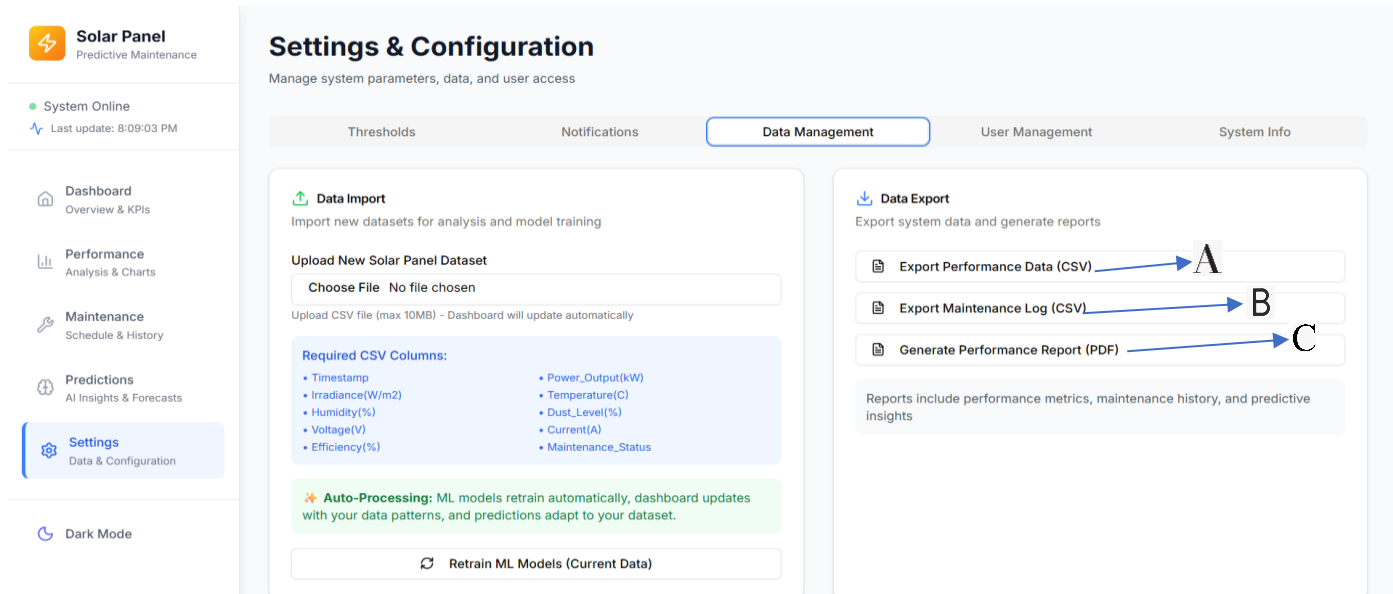


Fig.5.13. Data Management

A. Export Performance Data (CSV) Report

Table 5.1 Performance Data

	A	B	C	D	E	F	G	H	I	J
1	timestamp	power output	irradiance	temperature	humidity	dust level	voltage	current	efficiency	maintenance status
2	2020-05-14T18:00:00	541.7	1781.41	35.97	46.04	17.94	236.38	3	85.13	Required
3	2020-05-14T17:00:00	574.08	1848.85	32.58	26.72	24.16	231.14	1.765	90.49	Warning
4	2020-05-14T16:00:00	508.62	2048.11	34.4	34.69	26.11	235.21	3	78.53	Not Required
5	2020-05-14T15:00:00	437.73	1642.79	30.26	30.98	14.66	247.13	2.819	73.42	Not Required
6	2020-05-14T14:00:00	433.64	1931.82	37.79	40.28	13.58	236.83	3	92.49	Not Required
7	2020-05-14T13:00:00	414.31	1470.17	36.07	46.22	15.72	228.49	2.606	72.18	Required
8	2020-05-14T12:00:00	517.2	1903.89	30.91	55	19.83	223.51	1.856	86.39	Not Required
9	2020-05-14T11:00:00	398.71	1721.54	33.99	46.83	16.12	246.57	2.005	78.87	Not Required
10	2020-05-14T10:00:00	397.77	1624.55	39.46	28.31	3.32	239.95	2.714	84.55	Not Required
11	2020-05-14T09:00:00	563.11	1723	38.71	36.02	13.99	233.95	2.618	94.4	Not Required
12	2020-05-13T18:00:00	510.67	1855.18	27.34	37.89	12.94	232.83	2.577	89.36	Not Required
13	2020-05-13T17:00:00	428.89	1863.27	35.31	55	6.5	239.2	1.264	88.42	Not Required
14	2020-05-13T16:00:00	506.33	1699.73	30.1	37.92	12.63	233.7	2.892	85.21	Not Required
15	2020-05-13T15:00:00	371.74	1811.22	35.15	37.6	19.28	233.13	2.457	88.34	Warning
16	2020-05-13T14:00:00	485.11	1799.89	30.38	21.04	21.88	228.05	1.7	74.25	Required
17	2020-05-13T13:00:00	580	1746.62	33.63	48.89	14.45	230.75	2.422	83.99	Not Required
18	2020-05-13T12:00:00	388.61	2135.95	37.38	43.42	11.44	245.23	2.167	91.06	Not Required
19	2020-05-13T11:00:00	488.49	1873.27	33.16	38.4	5.42	239.79	2.393	86.75	Not Required
20	2020-05-13T10:00:00	600	1806.68	36.23	34.95	8.87	243.41	1.646	76.44	Not Required
21	2020-05-13T09:00:00	484.24	1996.37	37.26	41.12	18.7	239.51	2.831	94.44	Not Required
22	2020-05-12T18:00:00	585.76	1611.31	35.11	27.92	17.34	241.53	2.876	71.46	Not Required
23	2020-05-12T17:00:00	547.51	1500.65	34.19	48.97	5.12	231.9	2.232	70.52	Not Required
24	2020-05-12T16:00:00	387.77	1919.23	30.57	34.62	18.21	239.14	2.409	75.55	Not Required
25	2020-05-12T15:00:00	404.07	2019.32	36.57	46.07	15.63	236.47	2.019	73.52	Not Required
26	2020-05-12T14:00:00	424.04	1922.31	31.94	34.17	12.45	230.05	2.748	81.44	Not Required
27	2020-05-12T13:00:00	522.96	1693.87	36.99	35.97	11.36	232	2.221	76.68	Not Required
28	2020-05-12T12:00:00	451.76	2148.79	31.24	42.86	6.2	239.62	2.355	87.83	Not Required
29	2020-05-12T11:00:00	600	1655.85	37.18	39.63	23.63	229.89	1.761	83.36	Not Required
30	2020-05-12T10:00:00	530.49	1751.48	31.67	46.52	15.13	238.81	1.718	92.5	Not Required
31	2020-05-12T09:00:00	417.88	2020.86	35.84	43.86	17.17	233.28	2.14	82.14	Warning
32	2020-05-11T18:00:00	370.24	1687.26	36.25	32.31	17.66	230.88	2.708	80.96	Warning
33	2020-05-11T17:00:00	355.14	1726.08	30.86	47.91	26.78	229.75	2.581	93.95	Required
34	2020-05-11T16:00:00	371.5	1881.02	27.52	26.99	15.89	232.89	2.678	88.02	Not Required
35	2020-05-11T15:00:00	592.79	1735.49	39.7	28.83	18.32	229.53	1.927	94.2	Not Required
36	2020-05-11T14:00:00	488.34	1607.67	30.64	41.27	22.1	242.79	2.155	75.79	Warning
37	2020-05-11T13:00:00	460.93	1701.27	43.05	38.03	23.28	249.71	2.397	82.32	Not Required
38	2020-05-11T12:00:00	425.38	1969.83	33.4	41.44	18.72	225.6	1.868	88.84	Not Required

B. Export Maintenance Log (CSV)

The “Export Maintenance Log (CSV)” feature (Table 5.2) allows users to download all maintenance records, including task details, status, priority, and assigned technicians, in a structured spreadsheet format for further analysis and documentation. This enables easy tracking of historical maintenance activities, integration with external tools, and transparent reporting for audits or performance reviews.

Table. 5.2. Maintenance Logs in Excel

	A	B	C	D	E	F	G	H	I	J
1	panel_id	task_type	description	priority	scheduled_date	estimated_duration	assigned_to	status	created_at	completed_at
2	panel_A1	Repair	Repair the panel	High	2025-12-25T06:46:00+00:00	3 hrs	Shivkumar	Scheduled	2025-12-03T06:46:47.109884+00:00	
3	panel-9	Inspection	check the panel	Low	2025-11-29T08:01:00+00:00	1 hour	Siddharth	Scheduled	2025-11-27T08:01:53.749294+00:00	
4	panel-A7	Repair	Repair the panel	Medium	2025-12-03T17:52:00+00:00	4 hours	Shiv	Scheduled	2025-11-26T17:52:25.251206+00:00	
5	Panel-A5	Calibration	Caliberate the sensors	High	2025-11-28T17:49:00+00:00	5 hours	Sagar	Scheduled	2025-11-26T17:50:13.703630+00:00	
6	panel-c1	Replacement	replace panel	Medium	2025-11-27T11:09:00+00:00	1 hour	Moinu	Scheduled	2025-11-26T11:09:40.835892+00:00	
7	panel_b1	Inspection	check the panel	Low	2025-11-30T11:05:00+00:00	2 hours	Syed Ali	Scheduled	2025-11-26T11:05:21.968624+00:00	
8	Panel_A2	Cleaning	clean the panel	Medium	2025-11-27T06:24:00+00:00	1 hour	imam	Scheduled	2025-11-25T06:24:32.243183+00:00	
9										
10										

C. Generate Performance Report

The “Generate Performance Report” feature automatically compiles a detailed summary of the solar panel system’s recent operational data, including power output, efficiency, temperature, dust levels, and total records analyzed. It presents key metrics in structured tables, such as average, minimum, and maximum power and efficiency, along with the distribution of panel maintenance status. The report also includes recent time-stamped performance records, helping users quickly review abnormal readings or maintenance-required conditions. In addition, it provides visual analytics and forecast tables that highlight trends, expected future performance, and intelligent recommendations for maintenance and optimization.

The generated performance report is shown in Annexure – II.

5.3 MACHINE LEARNING IMPLEMENTATION

The Machine Learning implementation in this project uses models such as Random Forest Regressors and Classifiers, along with Isolation Forest, to predict power output and efficiency and to detect anomalies related to dust, temperature, shading, and other faults. These models are trained on historical sensor data and integrated into the backend so that the system can generate real-time predictions, maintenance forecasts, and AI-driven recommendations directly on the dashboard.

5.3.1 Data Processing Pipeline

Feature Engineering:

- Irradiance (W/m^2)
- Temperature ($^{\circ}\text{C}$)
- Humidity (%)
- Dust Level (%)
- Voltage (V)
- Current (A)
- Hour of day
- Day of year
- Month

5.3.2 Model Training Process

- Data Collection: Historical solar panel data from MongoDB
- Feature Extraction: Time-based and environmental features
- Data Preprocessing: Standard Scaler normalization
- Model Training: Random Forest algorithms (100 estimators)
- Model Evaluation: Train-test split (80-20)
- Model Persistence: Saved as .pkl files using joblib

5.3.3 Prediction Models

- **Efficiency Prediction Model**

Algorithm: Random Forest Regressor

Purpose: Predict future panel efficiency

Output: Efficiency percentage (0-100%)

- **Power output Model**

Algorithm: Random Forest Regressor

Purpose: Predict future panel efficiency

Output: Efficiency percentage (0-100%)

Maintenance Classification Model

Algorithm: Random Forest Classifier

Purpose: Classify maintenance requirements

Classes: Not Required (0), Warning (1), Required (2)

5.4 DATABASE SCHEMA

The database schema is designed in MongoDB with separate collections for performance data, maintenance tasks, users, alerts, and configuration settings, allowing clean separation of concerns and scalable data storage. Each collection stores structured documents with timestamps, sensor readings, prediction results, and status fields so that the system can efficiently query, visualize, and manage solar panel performance and maintenance information.

5.4.1 Solar data Collection:

- id: String (UUID)
- timestamp: Date Time
- power output: Float
- irradiance: Float
- temperature: Float
- humidity: Float4
- dust level: Float
- voltage: Float
- current: Float
- efficiency: Float
- maintenance status: String
- maintenance description: String

5.4.2 Users collection:

- id: String (UUID)
- name: String
- email: String

- role: String
- status: String
- created at: Date Time

5.4.3 Maintenance tasks Collection:

- id: String (UUID)
- panel id: String
- task type: String
- description: String
- priority: String
- scheduled date: Date Time
- estimated duration: String
- assigned to: String
- status: String
- created at: Date Time
- completed at: Date Time

5.5 API ENDPOINTS

- GET /api/dashboard/kpis - Retrieve current KPIs
- GET /api/dashboard/performance?days={n} - Get performance data for n days
- GET /api/dashboard/alerts - Fetch active maintenance alerts

5.5.1 Predictions Endpoints

- GET /api/predictions/forecast - Get 7-day ML predictions

5.5.2 Maintenance Endpoints

- GET /api/maintenance/tasks - List all maintenance tasks
- POST /api/maintenance/tasks - Create new maintenance task
- PUT /api/maintenance/tasks/{id} - Update task status
- DELETE /api/maintenance/tasks/{id} - Delete maintenance task

5.5.3 Data Management Endpoints

- POST /api/data/import - Import CSV data
- POST /api/data/train - Trigger model training

5.6 INSTALLATION AND DEPLOYMENT

Installation and deployment involve setting up the FastAPI backend with the required Python dependencies, configuring the MongoDB database, and loading the trained machine learning models for prediction services. The React frontend is then built and deployed, with environment variables configured so that the dashboard can securely communicate with the backend APIs and serve real-time monitoring, analytics, and maintenance features over the network.

5.6.1 Prerequisites

- Python: 3.8 or higher
- Node.js: 14.x or higher
- MongoDB: 4.4 or higher
- npm/yarn: Latest version

5.6.2 Backend Setup

Step 1: Navigate to backend directory
`cd backend`

Step 2: Install Python dependencies
`pip install -r requirements.txt`

Step 3: Start the backend server
`python -m uvicorn server:app --reload --host 0.0.0.0 --port 8000`

5.6.3 Frontend Setup

Step 1: Navigate to frontend directory
`cd frontend`

Step 2: Install Node.js dependencies
`npm install`

Step 3: Start the development server
`npm start`

5.7 KEY BENEFITS

The key benefits of this system include improved solar panel efficiency through continuous monitoring and early detection of performance degradation, dust accumulation, and thermal issues. It reduces unplanned downtime and maintenance costs by enabling predictive maintenance based on data-driven insights and anomaly detection. The solution also enhances decision-making by providing intuitive dashboards, reports, and forecasts that quantify energy losses, ROI, and maintenance priorities. Additionally, its scalable IoT–AI architecture makes it suitable for different plant sizes, supporting remote monitoring and long-term reliability of solar installations.

5.7.1 Operational Benefits

- **Reduced Downtime:** Predictive maintenance prevents unexpected failures
- **Cost Optimization:** Proactive maintenance reduces emergency repair costs
- **Extended Equipment Life:** Timely maintenance prolongs panel lifespan
- **Improved Efficiency:** Optimized cleaning and maintenance schedules
- **Data-Driven Decisions:** Analytics-based operational strategies

5.7.2 Technical Benefits

- **Real-time Monitoring:** Instant visibility into system performance
- **Automated Alerts:** Proactive notification of potential issues
- **Scalability:** MongoDB and FastAPI support large-scale deployments
- **Accuracy:** Machine learning improves prediction accuracy over time
- **Integration Ready:** RESTful API for third-party integrations

5.7.3 Business Benefits

- **ROI Improvement:** Maximized energy output and reduced costs
- **Compliance:** Automated maintenance logs for regulatory requirements
- **Reporting:** Comprehensive performance reports for stakeholders
- **Competitive Advantage:** Advanced analytics capabilities
- **Sustainability:** Optimized renewable energy production

6. PROTOTYPE MODEL DEVELOPMENT

The prototype model development involves building a scaled solar panel setup with integrated sensors, Arduino-based data acquisition, and Wi-Fi-enabled data logging to validate real-time monitoring and predictive maintenance concepts before large-scale deployment.

6.1 FLOW DIAGRAM OF THE MODEL

The flow of information in the prototype model can be summarized as the following flow diagram:

Solar Panel → Sensors (LM35, DHT11, ACS712, Voltage Sensor) → Arduino UNO → Serial Data (CSV string) → PLX-DAQ in Excel → Cleaned Dataset (CSV) → Machine Learning Model → Dashboard → Predictive Maintenance Decision

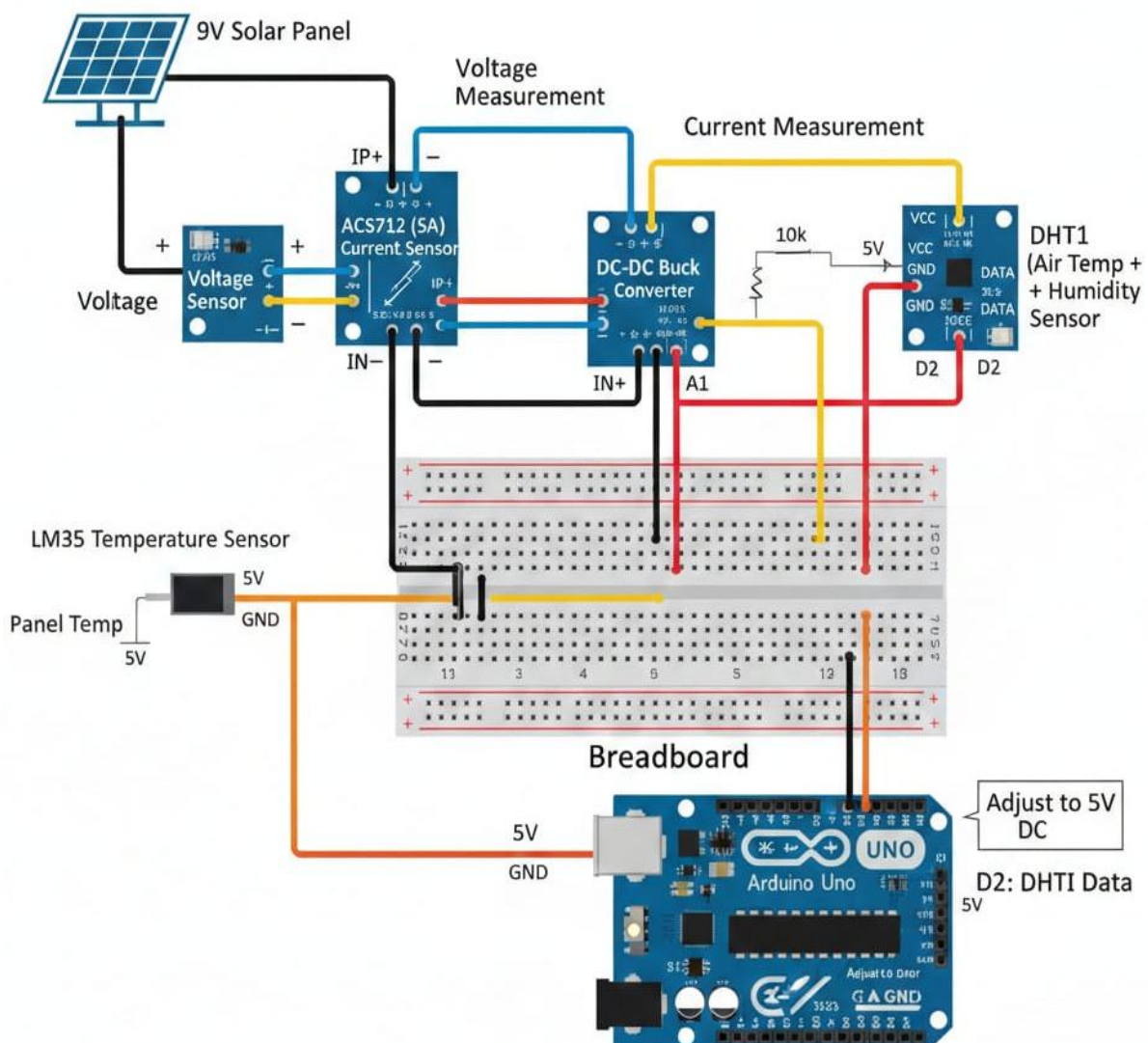


Fig 6.1. Circuit Diagram

6.2 DETAILED HARDWARE CONNECTIONS

Components and their connection are explained in detailed below.

6.2.1 Solar Panel

The miniature 9 V, 100 mA solar panel is the main energy source in the model. It has two output terminals: positive (+) and negative (-). For ease of wiring, wires are soldered to these copper pads and then connected to the breadboard. The panel positive line is routed both to the voltage sensor and through the ACS712 current sensor so that voltage and current can be measured simultaneously. The panel negative line is connected to the common ground rail on the breadboard.



Fig 6.2. 9V Solar panel

6.2.2 LM35 – Panel Temperature Sensor

The LM35 is a three-pin analog temperature sensor. When its flat face is kept towards the observer and the legs are pointing downwards, the pinout from left to right is: VCC, Vout, GND.

- VCC (left pin) is connected to the 5 V pin of the Arduino UNO.
- Vout (middle pin) is connected to analog input A0 of the Arduino.
- GND (right pin) is connected to Arduino GND.

The LM35 produces 10 mV per °C. The Arduino reads this analog voltage and converts it into panel surface temperature in degrees Celsius.

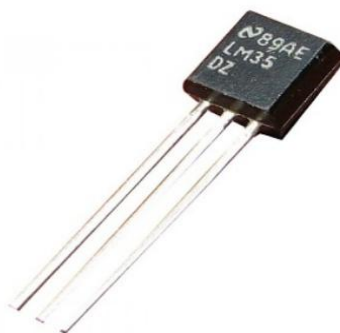


Fig 6.3. LM35

6.2.3 DHT11 – Ambient Temperature and Humidity Sensor

The DHT11 module used in the project has three pins labelled VCC, DATA and GND. It is used to measure the surrounding air temperature and relative humidity near the solar panel.

- VCC is connected to the 5 V pin of the Arduino.
- DATA is connected to digital pin D2 of the Arduino.
- GND is connected to Arduino GND.

The Adafruit DHT library is used in the Arduino code to read the digital data from the sensor and convert it to temperature (°C) and humidity (%).

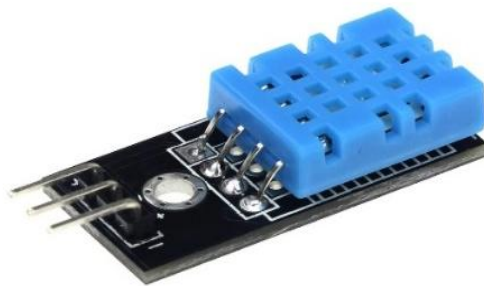


Fig 6.4. DHT11

6.2.4 ACS712 – Current Sensor

The ACS712 5 A current sensor measures the current produced by the solar panel. It has two high-current terminals (IP+ and IP–) for the current-carrying conductor, and three low-voltage pins: VCC, OUT and GND.

- The panel positive terminal is connected to IP+ of the ACS712.
- The load side (or measurement node) is connected to IP– of the ACS712.
- VCC is connected to the Arduino 5 V pin.
- GND is connected to Arduino GND.
- OUT is connected to analog input A1 of the Arduino



Fig 6.5. ACS712

6.2.5 DC Voltage Sensor Module

The DC voltage sensor is a simple resistive divider module that allows measurement of voltages higher than 5 V by scaling them down to the Arduino's safe range. It typically has pins labelled IN+, IN-, VCC, GND and OUT.

- IN+ is connected to the solar panel positive terminal.
- IN- is connected to the solar panel negative (ground).
- VCC is connected to the Arduino 5 V pin.
- GND is connected to Arduino GND.
- OUT is connected to analog input A2 of the Arduino.

The Arduino reads the OUT voltage and multiplies it by a fixed scale factor (VOLTAGE_RATIO) corresponding to the resistor ratio of the module to recover the actual panel voltage.



Fig.6.6. 25V DC Voltage Sensor

6.2.6 DC-to-DC Buck Converter

The **DC–DC buck converter** is used in our project to step down the **9V output from the solar panel to a stable 5V supply** required by the control and IoT components. It efficiently regulates voltage while minimizing power loss, ensuring reliable operation even with fluctuating solar input. The buck converter also protects sensitive electronics from overvoltage and improves overall system efficiency. This makes it suitable for continuous and stable power delivery in solar-based applications.



Fig.6.7. DC-to-DC Buck Converter

6.2.7 Arduino UNO and Laptop Connection

The Arduino UNO is powered and communicated with using a single USB cable connected to the laptop. Through this USB connection, the Arduino sends sensor readings as serial text data. The PLX-DAQ Excel macro reads this serial stream and logs it directly into an Excel sheet in real time.



Fig 6.8. Arduino Board

6.2.8 ESP 8266 or WI-FI Module

The **ESP module (ESP8266)** is used as the main IoT controller in our major project for wireless data communication. It collects sensor data and transmits real-time information to the server/dashboard using **Wi-Fi connectivity**. The module enables **remote monitoring and control** of the system through a web interface or cloud platform. Its low power consumption, built-in TCP/IP stack, and fast processing make it ideal for **real-time smart energy applications**.



Fig.6.9. WI-FI Module

6.3 ARDUINO FIRMWARE LOGIC

The Arduino sketch continuously reads sensor values, converts them into physical units and then sends them to the laptop in a PLX-DAQ compatible format. The key steps in the code are as follows:

- Initialize the serial port at 9600 baud.
- Initialize the DHT11 sensor.
- In setup(), send PLX-DAQ commands: CLEARDATA to clear old rows and LABEL to define column names.
- Inside loop(), read the LM35 analog value and convert it into panel temperature.
- Read air temperature and humidity from the DHT11 sensor.
- Read the ACS712 analog value, subtract the offset and divide by sensitivity to obtain current, then take the absolute value.
- Read the DC voltage sensor analog value and scale it with VOLTAGE_RATIO to obtain panel voltage.
- Calculate electrical power as $P = V \times I$.
- Compute elapsed time since startup using `millis() / 1000` to get time in seconds.
- Print a single line using the PLX-DAQ DATA format with all parameters separated by commas.
- Wait for a fixed sampling delay (e.g., 2 seconds) and repeat the loop.

6.4 DATA LOGGING AND FLOW INTO EXCEL

To avoid using Wi-Fi or cloud for the prototype, the project uses PLX-DAQ, an Excel macro tool that reads serial data from Arduino and logs it into an Excel spreadsheet in real time. The flow is:

- PLX-DAQ is opened in Excel (a .xlsm macro-enabled file).
- The correct COM port and baud rate (9600) are selected in the PLX-DAQ control panel.
- When the user clicks "Connect", the macro starts reading from the Arduino serial port.
- The Arduino sends one header line (LABEL,...) followed by many data lines (DATA,...).
- Each DATA line is parsed by PLX-DAQ and inserted into the next row in Excel.
- The resulting sheet contains a time-series dataset of all parameters including time, voltage, current, power, panel temperature, air temperature and humidity.

6.5 CONCEPTUAL FLOW OF THE COMPLETE PROJECT

The overall logical flow of the project can be described in the following stages:

- Stage 1 – Data Acquisition: Sensors continuously measure physical quantities from the solar panel environment.
- Stage 2 – Signal Conversion: The Arduino converts raw analog and digital readings into voltage, current, temperature, humidity and power.
- Stage 3 – Local Data Logging: The processed values are streamed via serial and captured in Excel using PLX-DAQ.
- Stage 4 – Dataset Preparation: Logged Excel data is cleaned, labelled with conditions (good, dusty, overheated, faulty) and exported as CSV.
- Stage 5 – Model Training: Machine learning algorithms such as Random Forest are trained on the CSV dataset to learn patterns between parameters and health conditions.
- Stage 6 – Dashboard Integration: The trained model is integrated into a dashboard application. Users upload new CSV files from the prototype, and the dashboard visualises graphs and predicts panel health.
- Stage 7 – Predictive Maintenance Decisions: Based on the model's output, the system suggests maintenance actions such as cleaning the panel, checking for overheating or investigating possible faults.

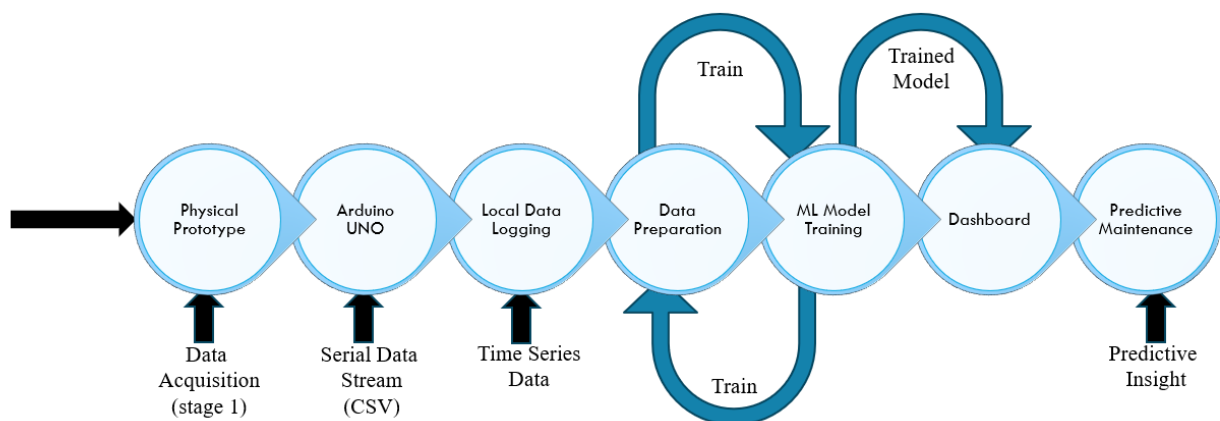


Fig. 6.10. Flow of the Complete Project

6.6 PROTOTYPE DEVELOPMENT

The prototype consists of a small-scale solar panel setup integrated with multiple sensors and an Arduino-based data acquisition system to emulate a real solar installation. It uses sensors such as ACS712 for current, LM35 for panel temperature, DHT11 for ambient temperature and humidity, a voltage sensor, irradiance sensor, and a dust sensor to capture all critical performance parameters. An LM2596 buck converter provides regulated power to the Arduino and Wi-Fi module, ensuring stable operation of the entire measurement circuit. The Arduino reads analog and digital signals, performs basic preprocessing, and sends the processed data to a PC or cloud through the Wi-Fi module at regular intervals. The prototype logs data into Excel using tools like PLX-DAQ, creating a live dataset for analysis, visualization, and model training. It demonstrates the complete data flow from sensor acquisition to power calculation, anomaly checks, and structured storage. The physical prototype also validates sensor placement, wiring, and noise-handling strategies that would be required in a larger solar plant. Overall, this prototype serves as a proof of concept for real-time monitoring and predictive maintenance, bridging hardware sensing with software analytics and dashboards.

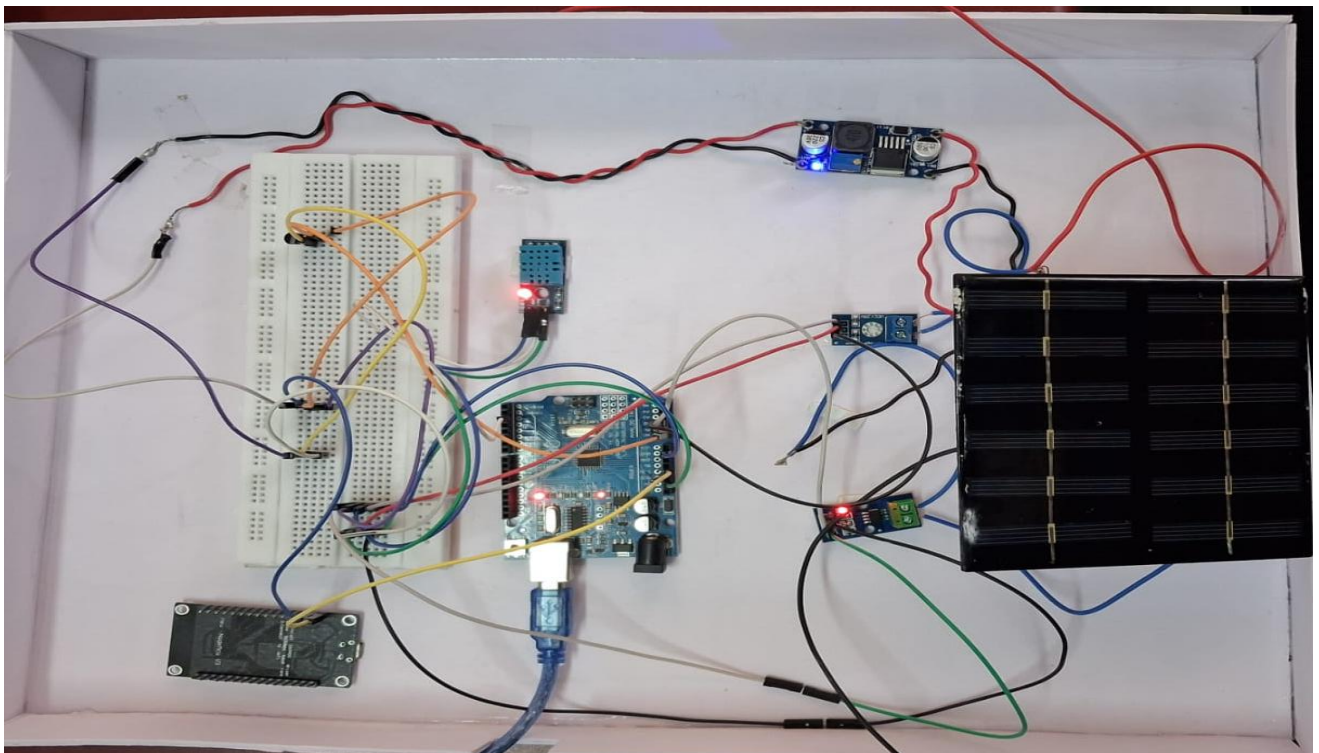


Fig.6.11. Prototype Model

7. CONCLUSION

The Solar Panel Performance Monitoring and Predictive System addresses the critical need for continuous monitoring and predictive maintenance of solar panel installations. Through comprehensive literature review and system design analysis, the project has established a clear framework for developing an intelligent monitoring solution.

The proposed system architecture incorporates IoT sensors, machine learning algorithms, and cloud-based analytics to enable real-time performance monitoring and fault prediction. The methodology encompasses hardware sensor integration, software development for data processing, and implementation of predictive analytics for maintenance optimization.

Current progress includes completion of literature review, system requirements definition, and initial architecture design. The research phase has identified key technologies and approaches that will be implemented in the upcoming prototype development phase. The project demonstrates significant potential for improving solar panel efficiency and reducing maintenance costs through proactive monitoring and early fault detection.

8. FUTURE SCOPE

Future enhancements include integrating thermal imaging or IR cameras to automatically detect hotspots, micro-cracks, and cell-level faults that are not captured by basic electrical sensors. The system can be extended with automated cleaning mechanisms or robotic cleaners triggered by dust level predictions and efficiency drops. Advanced deep learning models such as LSTMs or hybrid architectures may be incorporated to improve long-term forecasting accuracy under complex environmental conditions. Additionally, features like mobile app integration, multi-plant portfolio monitoring, and stronger cybersecurity measures can enhance usability, scalability, and secure remote access.

Planned Features

- Mobile Application: Android apps for remote monitoring
- Advanced Analytics: Deep learning models for improved predictions
- IoT Integration: Direct sensor data integration
- Weather API Integration: Real-time weather data for better predictions
- Multi-site Management: Support for multiple solar installations
- Automated Reporting: Scheduled email reports
- Anomaly Detection: AI-powered anomaly identification
- Energy Storage Integration: Battery system monitoring
- Cost Analysis Module: Financial impact analysis
- Predictive Alerts: SMS/Email notifications

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9. GANTT CHART

Task No.	Task	Plan/ Actual	Mar 25	Apr 25	May 25	Jun 25	Jul 25	Aug 25	Sep 25	Oct 25	Nov 25	Dec 25
1	Idea Selection	Plan										
		Actual										
2	Problem Definition	Plan										
		Actual										
3	Objectives	Plan										
		Actual										
4	Literature Review	Plan										
		Actual										
5	Methodology	Plan										
		Actual										
6	Data Collection	Plan										
		Actual										
7	Dashboard Development	Plan										
		Actual										
8	Model Development	Plan										
		Actual										
9	Testing	Plan										
		Actual										
10	Results and Conclusion	Plan										
		Actual										
11	Report Preparation	Plan										
		Actual										
12	CIE Presentation	Plan										
		Actual										
13	Final Presentation	Plan										
		Actual										

ANNEXURE – I

Arduino Code

```
#include <DHT.h>

// ===== Pin Definitions =====
#define LM35_PIN A0
#define ACS712_PIN A1
#define VOLTAGE_SENSOR_PIN A2

#define DHTPIN 2      // DHT DATA pin connected to D2
#define DHTTYPE DHT11 // Change to DHT22 if your sensor is DHT22

// ===== Objects =====
DHT dht(DHTPIN, DHTTYPE);

// ===== Calibration Values =====
// Adjust later if needed after testing
float ACS_OFFSET = 2.5;      // Midpoint voltage for ACS712 (approx)
float ACS_SENSITIVITY = 0.185; // V/A for ACS712 5A module (use 0.100 for 20A, 0.066 for 30A)
float VOLTAGE_RATIO = 5.0;   // Depends on your DC voltage sensor (often 5 or 25)

void setup() {
  Serial.begin(9600);      // Serial to PC / PLX-DAQ
  dht.begin();
  delay(2000);

  // === PLX-DAQ header ===
  Serial.println("CLEARDATA");
  Serial.println("LABEL,Time_s,PanelTemp_C,AirTemp_C,Humidity_pct,Voltage_V,Current_A,Power_W");
}

void loop() {
  // ===== 0. Time stamp (seconds since power on) =====
  float time_s = millis() / 1000.0;

  // ===== 1. LM35 Temperature (Panel Temp) =====
  int lm35_raw = analogRead(LM35_PIN);
  float lm35_voltage = (lm35_raw * 5.0) / 1023.0; // Convert ADC to voltage (0-5V)
  float panelTemp = lm35_voltage * 100.0;        // LM35: 10mV per °C

  // ===== 2. DHT Air Temp & Humidity =====
  float airTemp = dht.readTemperature(); // °C
  float humidity = dht.readHumidity();    // %

  // Handle possible read errors (NaN)
  if (isnan(airTemp) || isnan(humidity)) {
    airTemp = -1.0;
    humidity = -1.0;
  }
}
```

```

// ===== 3. ACS712 Current Sensor (always positive current) =====
int acs_raw = analogRead(ACS712_PIN);
float acs_voltage = (acs_raw * 5.0) / 1023.0;
float current = (acs_voltage - ACS_OFFSET) / ACS_SENSITIVITY; // Amps
current = abs(current); // force positive

// ===== 4. DC Voltage Sensor =====
int volt_raw = analogRead(VOLTAGE_SENSOR_PIN);
float sensor_voltage = (volt_raw * 5.0) / 1023.0; // Voltage at sensor OUT pin
float panelVoltage = sensor_voltage * VOLTAGE_RATIO; // Real panel voltage

// ===== 5. Power Calculation =====
float power = panelVoltage * current; // Watts (P = V x I)

// ===== 6. Send PLX-DAQ compatible DATA line =====
// Format: DATA,Time_s,PanelTemp_C,AirTemp_C,Humidity_pct,Voltage_V,Current_A,Power_W
Serial.print("DATA,");
Serial.print(time_s, 1);      Serial.print(",");
Serial.print(panelTemp, 2);   Serial.print(",");
Serial.print(airTemp, 2);     Serial.print(",");
Serial.print(humidity, 2);    Serial.print(",");
Serial.print(panelVoltage, 2); Serial.print(",");
Serial.print(current, 3);     Serial.print(",");
Serial.println(power, 3);

delay(2000); // Wait 2 seconds between samples
}

```

WI-FI Module Code

```
#include <ESP8266WiFi.h>

const char* ssid = "Imam";
const char* password = "1122334455";

void setup() {
    Serial.begin(115200);
    delay(1000);

    Serial.println();
    Serial.println("Starting WiFi...");

    WiFi.mode(WIFI_STA);
    WiFi.begin(ssid, password);

    int attempt = 0;
    while (WiFi.status() != WL_CONNECTED) {
        delay(1000);
        Serial.print("Connecting... Attempt ");
        Serial.println(attempt);
        attempt++;

        if (attempt > 15) {
            Serial.println("FAILED to connect.");
            break;
        }
    }

    if (WiFi.status() == WL_CONNECTED) {
        Serial.println("WiFi Connected!");
        Serial.print("IP Address: ");
        Serial.println(WiFi.localIP());
    }
}

void loop() {}
```

ANNEXURE – II

Solar Panel Performance Report

Generated on: 2025-12-14 20:50:37

Executive Summary

Metric	Value
Average Power Output	496.22 kW
Maximum Power Output	600.00 kW
Minimum Power Output	350.00 kW
Average Efficiency	83.64%
Average Temperature	34.79°C
Average Dust Level	14.51%
Total Records Analyzed	100

Panel Status Distribution

Status	Count
Not Required	81
Required	10
Warning	9

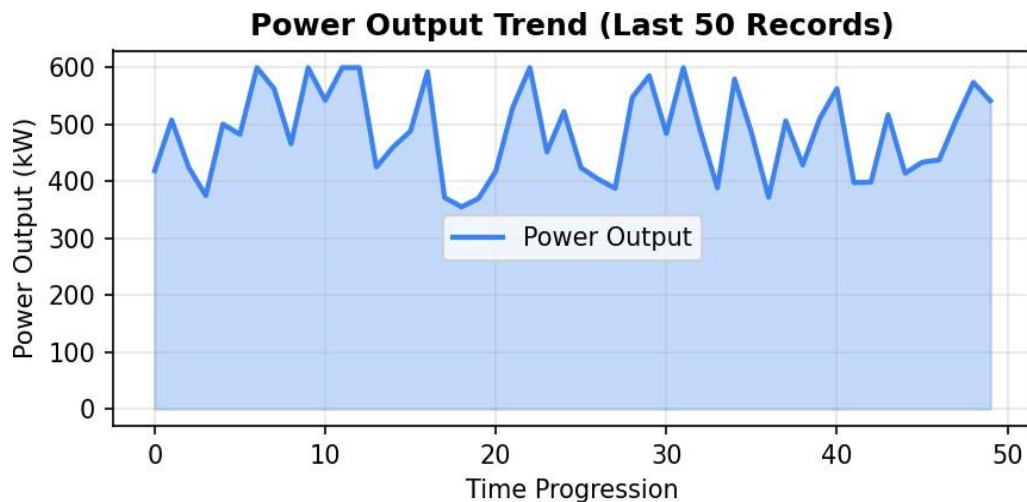
Recent Performance Data (Last 20 Records)

Timestamp	Power (kW)	Efficiency (%)	Temp (°C)	Status
2020-05-14 18:00	541.7	85.1	36.0	Required
2020-05-14 17:00	574.1	90.5	32.6	Warning
2020-05-14 16:00	508.6	78.5	34.4	Not Required
2020-05-14 15:00	437.7	73.4	30.3	Not Required
2020-05-14 14:00	433.6	92.5	37.8	Not Required
2020-05-14 13:00	414.3	72.2	36.1	Required
2020-05-14 12:00	517.2	86.4	30.9	Not Required
2020-05-14 11:00	398.7	78.9	34.0	Not Required
2020-05-14 10:00	397.8	84.5	39.5	Not Required

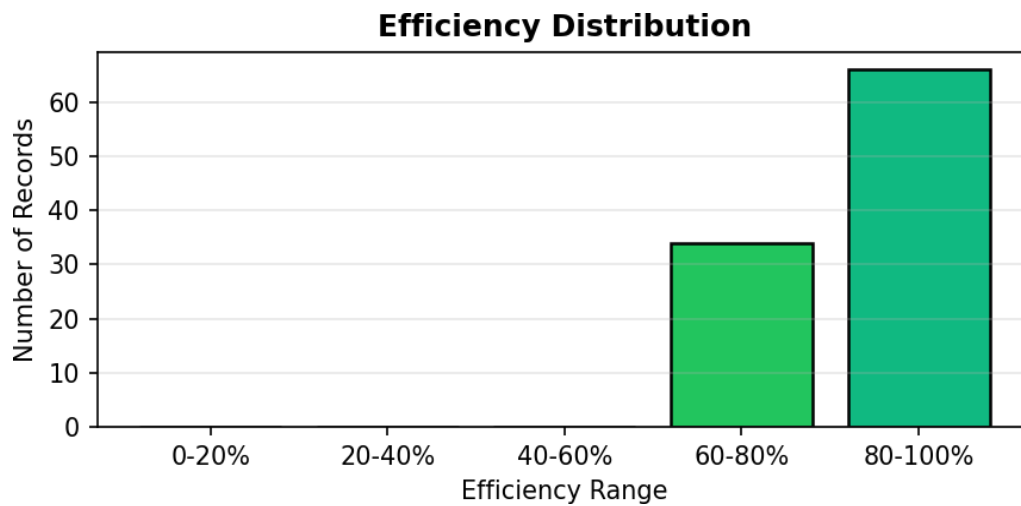
2020-05-14 09:00	563.1	94.4	38.7	Not Required
2020-05-13 18:00	510.7	89.4	27.3	Not Required
2020-05-13 17:00	428.9	88.4	35.3	Not Required
2020-05-13 16:00	506.3	85.2	30.1	Not Required
2020-05-13 15:00	371.7	88.3	35.1	Warning
2020-05-13 14:00	485.1	74.2	30.4	Required
2020-05-13 13:00	580.0	84.0	33.6	Not Required
2020-05-13 12:00	388.6	91.1	37.4	Not Required
2020-05-13 11:00	488.5	86.8	33.2	Not Required
2020-05-13 10:00	600.0	76.4	36.2	Not Required
2020-05-13 09:00	484.2	94.4	37.3	Not Required

Visual Analytics

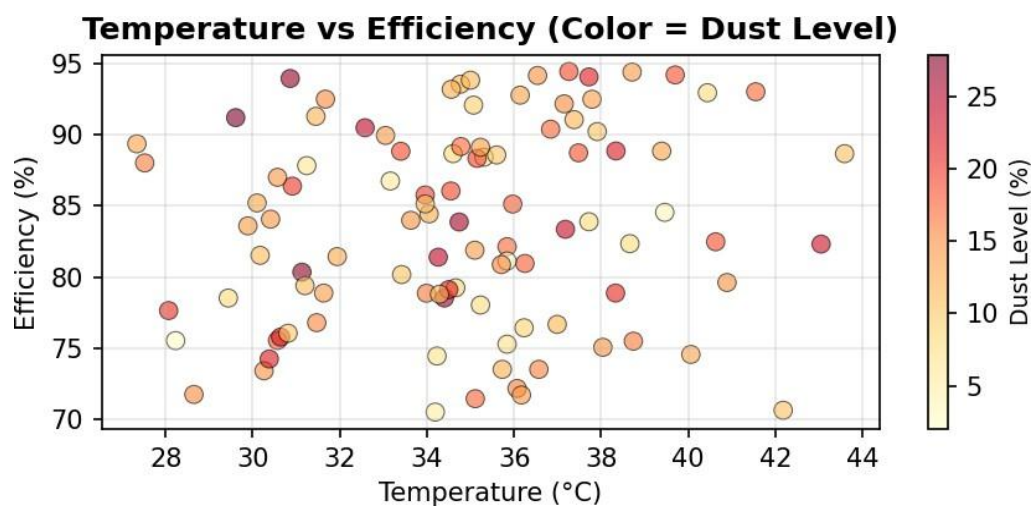
The following charts provide visual insights into your system's performance trends, efficiency distribution, and environmental factors affecting power generation.



Interpretation: This chart shows the power output trend over recent measurements. Upward trends indicate improving performance, while downward trends may signal degradation, dust accumulation, or environmental factors affecting generation.



Interpretation: This distribution shows how often your system operates at different efficiency levels. A healthy system should have most readings in the 60-100% range (green bars). High counts in lower ranges indicate potential issues requiring attention.



Interpretation: This scatter plot reveals the relationship between panel temperature and efficiency. Points are colored by dust level (yellow=low, red=high). Generally, higher temperatures reduce efficiency. Red points at high temperatures indicate combined stress from heat and dust, requiring immediate cleaning.

Predictions & Forecasts

This section provides 7-day forecasts for power output and efficiency based on historical patterns, environmental conditions, and machine learning models trained on your system's data.

Day	Predicted Power (kW)	Predicted Efficiency (%)	Confidence
Day 1	512.5	85.5	High
Day 2	513.7	82.3	High
Day 3	530.8	83.3	High
Day 4	526.1	82.0	Medium
Day 5	505.7	84.1	Medium
Day 6	507.8	82.3	Low
Day 7	514.9	81.2	Low

Intelligent Recommendations

Based on comprehensive analysis of your system's performance data, the following recommendations are generated to optimize efficiency, prevent failures, and reduce maintenance costs.

1. ■ GOOD: Dust levels are acceptable. Continue regular monitoring.
2. ■ EXCELLENT: System is operating at high efficiency.
3. ■ ATTENTION: 10 records flagged for maintenance. Review and schedule as needed.
4. ■ IMPROVING: Power output has increased by 10.8%. System performance is trending positively.

Maintenance Forecast

Panel Group	Current Status	Est. Days Until Maintenance	Priority
A-1	Good	65	Low
A-2	Good	65	Low
A-3	Good	65	Low
A-4	Good	65	Low
A-5	Good	65	Low
A-6	Good	65	Low
A-7	Good	65	Low
A-8	Good	65	Low
A-9	Good	65	Low
A-10	Good	65	Low

Efficiency Trend Analysis

Period	Average Efficiency (%)	Trend
Oldest Records	83.18	—
Middle Period	84.02	↑
Recent Records	83.41	↓

Overall System Health Score

Component	Score	Status
Efficiency Performance	92.9/100	✓
Dust Management	85.5/100	✓
Temperature Control	80.4/100	✓
Maintenance Status	90.0/100	✓
OVERALL HEALTH	87.2/100	✓

— End of Report —

Documentation Link

[https://github.com/ImamJamdar/Major Project](https://github.com/ImamJamdar/Major_Project)

Here we can find all the codes including dashboard and hardware Arduino code. Also, we can see the demonstration of the entire project.